

**COMPARATATIVE ASSESSMENT OF GOSAT,
GOSAT-2 AND TROPOMI FOR ATMOSPHERIC
METHANE (XCH₄) MONITORING OVER
MALAYSIA**

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GOSAT-2 AND TROPOMI FOR ATMOSPHERIC
METHANE (XCH₄) MONITORING OVER
MALAYSIA**

by

HENG WEI QUAN

**A dissertation submitted in partial fulfillment of the requirements for the degree of
Bachelor of Technology (B.Tech) in the field of Environmental Technology School
of Industrial Technology Universiti Sains Malaysia**

January 2026

DECLARATION BY AUTHOR

This dissertation is composed of my original work and contains no material previously published or written by another person except where due reference has been made in the text. The content of my dissertation is the result of work I have carried out since the commencement of my research project and does not include a substantial part of work that has been submitted to qualify for the award of any other degree or diploma in any university or other tertiary institution.



HENG WEI QUAN

January 2026

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LIST OF ABBREIATIONS

ITCZ	Intertropical Convergence Zone
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PENILAIAN PERBANDINGAN GOSAT, GOSAT-2 DAN TROPOMI BAGI PEMANTAUAN METANA ATMOSFERA (XCH_4) DI MALAYSIA

ABSTRAK

Pemantauan kepekatan metana atmosfera (XCH_4) di kawasan tropika seperti Malaysia menghadapi cabaran kritikal akibat litupan awan yang tebal serta kekurangan rangkaian stesen pemantauan darat. Kajian ini bertujuan untuk menilai keberkesanan teknologi penginderaan jauh satelit dengan membandingkan prestasi tiga platform berbeza: GOSAT, GOSAT-2, dan TROPOMI, bagi tahun 2019 dan 2024. Metodologi kajian melibatkan analisis taburan ruangan dan inter-komparasi statistik menggunakan perisian R Studio untuk mengukur ketepatan perolehan data dan mengenalpasti bias sistematik. Hasil kajian mendapati bahawa walaupun TROPOMI menawarkan resolusi ruangan yang tinggi dan mampu memetakan sumber pelepasan metana dengan jelas, instrumen ini mempamerkan ketidakstabilan radiometrik yang ketara, terutamanya ralat "bias albedo" di kawasan hutan tebal. Sebaliknya, GOSAT-2 menunjukkan kestabilan data yang tinggi dengan sisihan piawai yang rendah, hasil daripada teknologi "penunjuk pintar" yang mengurangkan gangguan awan, menjadikannya rujukan penentuan yang ideal. Analisis bermusim turut mengesahkan bahawa Monsun Timur Laut menghalang perolehan data yang sah secara ketara, manakala Monsun Barat Daya menawarkan tettingkap cerapan yang optimum. Oleh itu, kajian ini mencadangkan satu "Protokol Pemantauan Hibrid" yang mengintegrasikan penataan ruangan daripada TROPOMI dengan

penentuan ketepatan daripada GOSAT-2. Pendekatan ini menyediakan rangka kerja saintifik yang lebih jitu untuk inventori gas rumah hijau kebangsaan, mengatasi batasan yang wujud apabila bergantung kepada penderita tunggal.

**COMPARATIVE ASSESSMENT OF GOSAT, GOSAT-2 AND TROPOMI
FOR ATMOSPHERIC METHANE (XCH₄) MONITORING OVER
MALAYSIA**

ABSTRACT

Monitoring atmospheric methane concentrations (XCH₄) in tropical regions like Malaysia faces significant challenges due to persistent cloud cover and a lack of ground-based monitoring stations. This study aims to evaluate the efficacy of satellite remote sensing by comparing the performance of three distinct platforms: GOSAT, GOSAT-2, and TROPOMI, for the years 2019 and 2024. The methodology involved spatial distribution analysis and statistical inter-comparison using R Studio software to quantify retrieval accuracy and identify systematic bias. The results revealed that while TROPOMI offers high spatial resolution capable of mapping methane emission sources clearly, it exhibits significant radiometric instability, particularly "albedo bias" errors over dense rainforests. In contrast, GOSAT-2 demonstrated high data stability with a low standard deviation, attributed to its "intelligent pointing" technology which mitigates cloud interference, making it an ideal calibration reference. Seasonal analysis further confirmed that the Northeast Monsoon critically impedes valid data retrieval, whereas the Southwest Monsoon offers the optimal observation window. Consequently, this study proposes a "Hybrid Monitoring Protocol" that integrates the spatial mapping capabilities of TROPOMI with the precision calibration of GOSAT-2. This approach provides a more accurate scientific

framework for the national greenhouse gas inventory, overcoming the limitations inherent in relying on a single sensor.

CHAPTER 1

INTRODUCTION

1.1 Background of Study

One of the most pressing environmental issues of the 21st Century is the rise of anthropogenic (human-made) greenhouse gases (GHG) in the atmosphere as a result of climate change occurring globally. Of the many types of GHGs, methane (CH₄) poses a major concern because it has a global warming potential (GWP) that is estimated to be between 28 and 34 times more than carbon dioxide (CO₂) over a period of 100 years (IPCC, 2021). Thus, accurate monitoring and measuring of atmospheric methane levels is critical for ensuring that a country's reporting of its methane emissions complies with the requirements set forth by international agreements such as the Paris Agreement, and is needed to develop an effective strategy to mitigate these effects.

Ground observations of greenhouse gases (GHGs) have been the traditional source for GHG monitoring; however, ground infrastructures in tropical developing countries, such as Malaysia, are relatively scarce and limited in geographic coverage. Consequently, large portions of the Malaysian territory remain unmonitored like remote rainforest and agricultural areas. Additionally, recent developments in satellite-based remote sensing have provided an additional means to monitor atmospheric composition for large regions and on a global scale. The launch of the GOSAT satellite in 2009 was a significant step towards this goal. The GOSAT includes a Thermal And Near-infrared Sensor for carbon Observations (TANSO-FTS), which provides high accuracy,

point-based measurements of methane (CH_4). The GOSAT-2 satellite was launched in 2018 and included several innovative technologies like "Intelligent Pointing" technologies designed to help improve the accuracy of measurements taken through cloudy conditions.

The launch of the TROPOspheric Monitoring Instrument (TROPOMI) aboard the Sentinel-5 Precursor satellite in 2017 has changed the way we monitor methane emissions with high-resolution imaging of methane plumes. The TROPOMI system gives researchers daily continuous maps of methane concentration instead of the infrequent point measurements provided by the GOSAT series. Using optical sensors to monitor methane in humid tropical regions presents difficulties, as demonstrated in Malaysia. Because of its geographical position, Malaysia experiences a great deal of cloud cover and highly variable aerosol loading, which negatively impacts the precision of satellite measurements (Tangang et al., 2012). Consequently, a comparison of these various satellite technologies must be done to determine which ones are best suited for use at the national scale for monitoring the methane emission from Malaysia.

1.2 Problem Statement

Although satellite remote sensing may offer a feasible method for observing atmospheric methane concentrations in Malaysia, the present-day capability of remote sensing sensors in tropical climates is uncertain. One of the biggest challenges with the current systems is finding the right balance between the spatial coverage of the sensor and its radiometric accuracy. The legacy GOSAT sensor has been used as a reference for

high-precision measurements, but it provides extremely limited spatial coverage; consequently, many areas of Malaysia will be unmeasured using GOSAT alone. Conversely, while the TROPOMI sensor provides sufficient spatial resolution to identify possible sources of emissions, some initial findings indicate that the "Full-Physics" retrieval method used by TROPOMI is prone to errors caused by differences in albedo across various surfaces, most notably dark tropical trees and forests (Lorente et al., 2021).

Malaysia experiences strong meteorological patterns during both of the Monsoons, which limits the time for collecting usable data. Shortwave Infrared (SWIR) sensors are unable to penetrate through the heavy cloud cover created during the wet seasons, making data collected during the wet seasons severely interrupted and possibly incorrect. Currently, there is no agreed-upon national policy regarding what satellite sensor is best to use or how to integrate the many data sources from GOSAT and TROPOMI into a single national inventory of methane emissions. With reliance on a single sensor, without the seasonal limitations and biases from other instruments taken into account, there could be substantial errors in the reported emissions of methane in Malaysia. Therefore, there is an urgent need to methodically compare and validate these different satellite platforms in order to develop a scientifically reliable monitoring system.

1.3 Research Objectives

The primary aim of this study is to evaluate the performance of satellite remote sensing technologies in monitoring atmospheric methane over Malaysia. To achieve this, the following specific objectives have been established:

1. To evaluate the statistical distribution and seasonal availability of atmospheric methane (XCH_4) concentrations over Malaysia for the years 2019 and 2024, comparing the data density and coverage of GOSAT, GOSAT-2, and TROPOMI.
2. To assess the retrieval accuracy and systematic bias of TROPOMI and GOSAT-2 by performing a pairwise statistical inter-comparison (using Pearson Correlation and Root Mean Square Error) against the legacy GOSAT data as the reference standard.
3. To determine the most suitable satellite sensor for national monitoring by analyzing the impact of the Northeast and Southwest Monsoons on valid data retrieval rates, thereby establishing a protocol for selecting valid "true" methane data for Malaysia.

CHAPTER 2

LITERATURE REVIEW

In this chapter, a thorough review of the literature that has been published on monitoring atmospheric methane (CH_4) using remote sensing from space and the satellites that operate by utilizing this technology is presented. The beginning of this document will cover an overview of the role that methane plays in climate change on a global level, as well as the need to record methane emissions accurately on a national level. Following this, the fundamental physical processes that are used in remote sensing to retrieve gas information through infrared spectroscopy will be outlined. After that, three datasets of satellite measurements obtained through GOSAT, GOSAT-2, and TROPOMI will be critically analysed with respect to their respective designs and observational techniques. Finally, the retrieval methodologies and algorithms such as Proxy and Full-Physics that the remote sensing instruments have been utilizing, along with the problems that these instruments face in deriving these measurements from a tropical region, will be discussed, including the effects of cloud interference and inaccuracies caused by alluvial and surface properties.

2.1 Atmospheric Methane and Climate Change

Aside from carbon dioxide (CO_2), methane is the second most common greenhouse gas caused by humans that contributes to approximately 20% of the observed increase in global temperatures from pre-industrial times. Although the level of methane in the atmosphere is much lower than for CO_2 , it has a much greater potential

for Global Warming . The Intergovernmental Panel on Climate Change (IPCC) reports that methane's potential for Global Warming (GWP) is approximately 27–30 over the next 100 years, while it is over 80 for the next 20 years (IPCC 2021). Therefore, reducing methane emissions is the primary focus of immediate climate action.

In Southeast Asia, primarily, methane emissions are generated by both biogenic and pyrogenic processes. The main sources of methane emissions are anaerobic decomposition of organic matter in wetlands and paddy fields, enteric fermentation from livestock, and burning of biomass to clear land for agricultural production (Worden et al., 2022). The oil and gas industry is also an important source of fugitive emissions. Nevertheless, due to limited ground-based observation of methane emissions in tropical regions, accurately calculating the level of these emissions is problematic, resulting in large uncertainties regarding the global budget for methane emissions (Qu et al., 2021).

2.2 Principles of Satellite Remote Sensing of Methane

The determination of atmospheric methane concentration with satellite remote sensing is based on the interaction of methane molecules with solar radiation in the shortwave infrared (SWIR). Solar radiation is absorbed at specific wavelengths by methane, resulting in distinct “fingerprint” absorption features in the reflected sunlight spectrum. Satellite sensors measure the radiance of sunlight that has been reflected from the Earth’s surface; by analysing the depth of the absorption features in the 1.65 μm and 2.3 μm bands, it is possible to derive the total methane column density along the optical path (Hu et al., 2018). The column-averaged, dry air mole fraction, or XCH_4 , is the

primary measured product, which is defined as the ratio of the number of methane molecules to the number of molecules of dry air in a vertical column through the atmosphere. To achieve a precision of less than 1% (10 to 20 ppb for climate monitoring), the sensor must have both high spectral resolution and high signal-to-noise ratios. Retrieval of the methane column relies on having clear skies as clouds absorb SWIR radiation, blocking it from reaching the lower troposphere where most of the methane sources are located.

2.3 Satellite Missions for Methane Monitoring

In the past 10 years, the ability to observe methane emissions from space has developed through three different satellite instrument generations. This paper will examine satellite instruments within both the GOSAT series and the TROPOMI satellite instrument in order to assess their individual contributions to methane emissions measurement.

2.3.1 Greenhouse Gases Observing Satellite (GOSAT)

In 2009, JAXA launched GOSAT, or "IBUKI," as the first satellite to measure greenhouse gases on a global scale. It includes a Fourier Transform Spectrometer called TANSO-FTS and has a point mode which enables it to take measurements at individual points with a footprint of about 10.5 km in diameter, while separating the measurements by hundreds of km. Although the GOSAT data provided the first accurate baseline of methane on a global scale, due to its sparse sampling method, it is not suitable for identifying localized sources or constructing continuous maps of methane emissions (Parker et al., 2020). Nevertheless, because of its long service and validation against

ground-based TCCON stations, GOSAT data remains the reference for newer sensors in terms of cross-calibration.

2.3.2 GOSAT-2 (IBUKI-2)

GOSAT-2, which was launched in December 2018, is the newest instrument in the GOSAT series and is equipped with a TANSO-FTS-2 instrument; which focuses on improving upon the many limitations of the original GOSAT, and is also significantly more accurate than the original GOSAT instrument. One of the most important improvements over GOSAT is that GOSAT-2 has the ability to "intelligently" point towards the Earth for the most accurate observations possible. GOSAT-2 uses an onboard camera to identify the presence of clouds in the area being imaged in order to determine which areas are clear and therefore best suited for imaging by the TANSO-FTS-2 instrument. Because of this improvement in its design, the GOSAT-2 satellite is able to produce significantly more valid retrievals from data collected in areas affected by persistent cloud cover than was the case with the original GOSAT satellite. GOSAT-2 maintains the GOSAT legacy of providing high spectral precision and significantly increased valid retrieval yields in challenging atmospheric conditions.

2.3.3 TROPOMI (Sentinel-5 Precursor)

Launch of the Sentinel-5 Precursor satellite in 2017 with the TROPOMI instrument represented a dramatic departure from the point measurements associated with the GOSAT series. By employing a push-broom imaging spectrometer with a wide swath of 2,600 km, TROPOMI allows near-global daily coverage of the planet with an extremely high spatial resolution (7 km x 5.5 km). The ability to "map" methane

distribution as opposed to simply "sample" it, provides for the ability to identify emission hotspots and spatial gradients (Lorente et al., 2021). While the increased spatial resolution is associated with increased susceptibility to both instrumental noise and surface artifacts, TROPOMI is less susceptible to these influences compared to the high-precision FTS sensors on GOSAT.

2.4 Retrieval Algorithms and Challenges

Accurate derivation of methane concentrations from satellite radiance spectra involves using highly complex mathematical inversion algorithms. Most of the time, there are two standard approaches to achieving this goal: the Proxy Method and the Full-Physics approach.

2.4.1 Proxy vs. Full-Physics Methods

The Proxy Method was developed initially for retrieving GOSAT methane data; it determines the amount of methane CH_4 to carbon dioxide CO_2 ratio and then multiplies that by an atmospheric model of CO_2 concentrations. Since atmospheric scattering effects caused by clouds and aerosols occur close together in the spectrum for both methane and carbon dioxide, when calculating the ratio of the two, the cancelling effect of those atmospheric substances makes the Proxy Method one of the most robust and stable methods compared to other techniques, especially in instances of thin cloud coverage (Parker et al. 2020).

In contrast, the Full-Physics Method that is used by TROPOMI and GOSAT-2

attempts to explicitly model the scattering properties of aerosols and clouds and the concentration of gases simultaneously. The benefit to this approach is that it will allow one to retrieve methane independently from the CO₂ model. However, because the full-physics method is computationally intensive, it has a high level of sensitivity to any errors associated with the available readings used to develop the scattering model. Therefore, if the surface reflection (albedo) or aerosol load is mischaracterized, this could create large biases in the final XCH₄ product.

2.4.2 Challenges in Tropical Environments

Methane measurement in existing tropical locations, such as Malaysia, presents challenges due mainly to the presence of dense, cloudy skies. According to (Tangang et al. 2012), the ITCZ and Northeast Monsoons produce consecutive months of cloud cover that inhibit satellite-level observation or mock of the Earth.

Additionally, the dense vegetative cover of tropical rainforests can create a phenomenon called "albedo bias." The low reflectivity of dense vegetation in the range of SWIR creates a low signal-to-noise ratio for the satellite sensor. The latest studies show that Full-Physics Analysis Algorithms tend to misidentify the signal of the dark surface area as an image generated by the surrounding atmosphere, which has resulted in adverse systematic errors or "biases" in the estimated Methane concentrations measured by tropical environments (Lorente et al., 2021). Therefore, it is essential to conduct cross-validation of smaller maps and large-precision reference satellites to provide assurances of the accuracy and quality of this information.

CHAPTER 3

METHODOLOGY

This chapter describes the method used to evaluate variability and precision of atmospheric methane (XCH_4) concentrations over Malaysia. The method was specifically designed to meet the limitations of satellite remote sensing in tropical areas and uses a multi-platform technique which combines information from the GOSAT, GOSAT-2, and TROPOMI satellites. The analysis of the data is organised into four main stages, namely, Data acquisition, Data pre-processing and Quality Control, Spatiotemporal Co-location (matching), and Statistical analysis. The statistical computing and data processing of all stages of the procedure were conducted using the R programming language and R Studio. Thus, allowing for a reproducible and robust analysis pipeline.

3.1 Study Area

Geographically, this study encompasses the entire sovereign nation of Malaysia including Peninsular Malaysia which is West Malaysia and East Malaysia's Borneo-based states of Sarawak and Sabah. This study area in Malaysia is located near the equator at latitude 0°N to 8°N and longitude 99°E to 120°E . It has an equatorial humid tropical climate, has many challenges associated with passive satellite remote sensing due to excessive cloud cover, high humidity, and large concentrations of atmospheric aerosols. The occurrence of high levels of airborne moisture combined with a large number of aerosol particles in the air dramatically impacts the amount of solar

radiation that is transmitted through the Shortwave Infrared (SWIR) bands used to retrieve methane. Because of this unique set of atmospheric conditions, selecting this area of Malaysia represents an optimal location for evaluating valid rates of data retrieval from various sensor technologies that are affected by these different meteorological regimes, as represented by the Northeast Monsoon which is from November to March and Southwest Monsoon which is from May to September.

3.2 Data Acquisition

Databases were used to conduct a comparison of different generations of satellites. Three different databases were obtained for use in conducting this analysis, factoring in two different years which are 2019 (GOSAT) and 2024 (GOSAT-2). The database for both GOSAT and GOSAT-2 was obtained from the GOSAT Data Archive Service (GADAS) maintained by the National Institute for Environmental Studies (NIES). The analysis of 2019 data utilized the GOSAT Level 2 XCH₄ product that was created by using the TANSO-FTS. Data for 2024 was obtained from the advanced TANSO-FTS-2 instrument on board GOSAT-2, which provided the Level 2 data products by means of a "Full Physics" retrieval algorithm that employed the use of the satellite's "intelligent pointing" system aimed at increasing the number of valid observations by actively targeting cloud-free regions.

Simultaneously with the previous work, methane measurements with TROPOMI on the Sentinel-5 Precursor satellite were obtained through the Copernicus Open Access Hub. Both periods of time for this work utilized the Level 2 Methane (L2_CH4) product.

Rather than taking measurements of single point sources like the satellites of the Japanese GOSAT series, TROPOMI has daily near-global coverage with a substantially finer spatial resolution of about $7 \text{ km} \times 5.5 \text{ km}$. This capability for wide-swath imaging is necessary for the spatial mapping component of the analysis as it allows for the identification of methane plumes that would have been missed due to GOSAT's lower resolution.

3.3 Data Pre-processing and Quality Control

Rigorous filtering of raw satellite data was performed to produce a final analysis that only utilizes high-quality retrievals. For the TROPOMI dataset, a thorough filtering process was implemented based on the product's quality assurance value (`qa_value`) within its metadata. Per the recommended procedures by the European Space Agency, only pixels with `qa_value > 0.5` were included in this study; the selected threshold effectively removes all data points that are impacted by clouds, cirrus clouds and snow or ice, and retrievals with solar zenith angles > 70 degrees. Therefore, only cloud-free and true spatial maps were produced. The same process was done for both the GOSAT and GOSAT-2 datasets, as these two products have pre-processing techniques using the validity flags associated with the SWIR spectra prior to co-locating the data points. Additional filtering was employed on the data points based on aerosol optical thickness to eliminate scenes that have excessive scattering that could create errors in concentration measurements. The result of this pre-processing phase is a set of valid methane retrieval points for 2019 and 2024 in preparation for co-location analysis.

3.4 Spatiotemporal Co-location Strategy

An in-house spatio-temporal co-location algorithm has been built and executed with R Studio for the validation of both the pairwise statistical intercomparisons of all three satellite sensors as well as through the use of an algorithm paired with triplet validation events. Due to the various orbits and relative overpass times of each satellite sensor, it would be highly improbable for them to ever perfectly coincide in time or space. To that end, a set of criteria was established to determine eligible sensor pair matches. Therefore, an eligible match shall be found when a TROPOMI pixel's centre and a GOSAT or GOSAT-2 footprint's centre are separated by less than 0.1 degrees (approximately 11 km). By setting such a close proximity limit ensures that both satellite sensors are viewing the same atmospheric column. Additionally, because the satellite observations are taken from the same calendar day, the spatio-temporal algorithm assumes that this is a proper baseline for validating the background data. Given the relatively long lifetime of methane in the atmosphere, daily data associations will suffice as a basis for validation. However, the algorithm assumes that no major transport of local plumes will occur between TROPOMI (approximately 13:00 local time) and GOSAT series data (approximately 13:30). The spatio-temporal algorithm has been designed to iterate across the full set of datasets to produce pair matches for the following sets which are match pairs of TROPOMI and GOSAT, TROPOMI and GOSAT-2, GOSAT and GOSAT-2, and all triplet coincidences.

3.5 Statistical Analysis

A thorough statistical evaluation was conducted looking at how accurately the co-located datasets were retrieved and how much bias there might be due to a systematic error in their retrieval. As part of this evaluation, we calculated the Mean Bias to measure the systematic difference between each sensor's readings and the reference standard sensor's readings. The Mean Bias shows us whether each sensor reads higher or lower 'on average' than the reference sensor for each of the methane concentrations that were tested. If the Mean Bias is positive, that means the sensor over-replicates; if it is negative, then it under-replicates.

After conducting a bias assessment, Root Mean Square Error (RMSE) was calculated to measure random error (variance) and how far away from the best-fit regression line (predicted value) the actual data points fall. RMSE provides insight into the accuracy of measurements, where lower RMSE values indicate a closer relationship between measurements. Additionally, the Pearson Correlation Coefficient (r) was calculated to indicate how closely these two data sets are related. By calculating the correlation coefficient, it is possible to compare how well time trend lines and seasonal variations of methane concentration are captured by each sensor. Collectively, these three statistical measurements provide quantitative information regarding accuracy, which will be presented in the results and discussion section of the thesis.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter encompasses a thorough examination of atmospheric methane (XCH_4) levels based on GOSAT, GOSAT-2 and TROPOMI satellite measurements taken between 2019 and 2024. The study will be divided into three main categories: (1) Statistical variability and seasonal distribution of the methane data sets, (2) Estimate of the retrieval accuracy and systematic errors through direct comparisons between the three satellite instruments, and (3) Establishment of which satellite instrument is most suitable for monitoring the Malaysian area by assessing the effects of the Monsoon season on data returns. The analysis will first consist of a spatial analysis of data coverage from all three satellites providing insight into the types of sampling method utilized by each satellite; this will include comparisons between the point-based sampling from the GOSAT series, versus the imaging capabilities provided by TROPOMI. Secondly, the validation process will quantitatively validate the sensors based on comparisons to legacy GOSAT data from 2019 and eventually include TROPOMI and GOSAT-2 comparisons by 2024. Lastly, the study assesses how the Northeast and Southwest Monsoons influence retrieval rates to establish a robust protocol for the selection of valid methane data for Malaysia.

4.1 Spatiotemporal Analysis of Methane Concentration (XCH_4) in 2019 and 2024

The current study provides a detailed and intricate analysis of the temporal variations in the amount of Atmospheric Methane (XCH_4) present in Malaysia including

the Peninsular Malayan states of Sarawak and Sabah during two non-consecutive calendar years, 2019 and 2024. Satellite observations taken on board three different satellite platforms which are TROPOMI (Sentinel-5P), GOSAT (TANSO-FTS) and GOSAT-2 (TANSO-FTS-2) provide complete monthly dataset for the entire period of observation January to December in 2019 and 2024. The analyses present a detailed chronological record of the effect of both the Northeast and Southwest Monsoon on seasonal variations in the atmospheric concentrations of methane across the Malaysian region, as well as determine the location of significant sources of methane emissions throughout Malaysia.

4.1.1 TROPOMI (Sentinel-5P) Observations

In the year 2019, the TROPOMI sensor provided high-resolution satellite images to map the distributions of methane across the country but was limited by seasonal cloud coverage through much of the year. The overall average concentration throughout Malaysia for the year was found to be 1854.67 ppb (parts per billion). The beginning of the year had minimal data coverage due to the presence of the Northeast Monsoon which January had 1846.85 ppb average monthly concentration. There were considerable gaps in satellite imagery during this time frame due to excessive cloud density and, therefore, only valid data could be obtained from around the coastal areas of Peninsular Malaysia. As the Northeast Monsoon continued through February, the average concentration for Peninsular Malaysia increased minimally compared to January which is 1849.23 ppb average for the month. However, coverage remained fragmented throughout the month

as well. The transitional month of March was expected to produce a slight decline in the average concentration which is 1839.45 ppb, possibly due to atmospheric mixing characteristics associated with the transition from the Northeast Monsoon to the Southwest Monsoon.

In April, with weather changes, the amount of area covered increased greatly and average concentrations reached 1843.12 ppb which improved detection of emission plumes from metropolitan regions. With the arrival of dry air from the Southwest Monsoon in May extensive data existed with a monthly average of 1852.67 ppb. June had the highest quality data and average concentrations of 1845.89 ppb which enabled local hotspot mapping with high resolution. The quality continued through July with a concentration of 1843.56 ppb showing background concentrations were fairly consistent through the central portion of the forest.

In August, although regional haze episodes were starting to impact optical retrievals, the average concentration remained quite consistent with an average of 1855.34 ppb for this month. September had one of the clearest spatial distribution maps of all the months in the year with a mean value of 1855.12 ppb for the entire month. The large amount of retrievals during this month gave the opportunity for a detailed evaluation of emissions throughout the Borneo region, which would have allowed for the examination of potential biases due to albedo over the extensive areas of densely populated forests. In October the return of inter-monsoon resulted in increased cloud coverage and the mean level of Methane increased to 1866.78 ppb. The beginning of the

Northeast Monsoon in November again resulted in reduced retrievals from the non-monitored region and caused an increase in the mean level of Methane to 1857.45 ppb. The final month of December had a small amount of valid data available due to extensive cloud cover. However, the mean level of methane based on the remaining pixels indicates 1863.23 ppb, which would represent an accumulation in the overall atmospheric level of methane throughout the entire year.

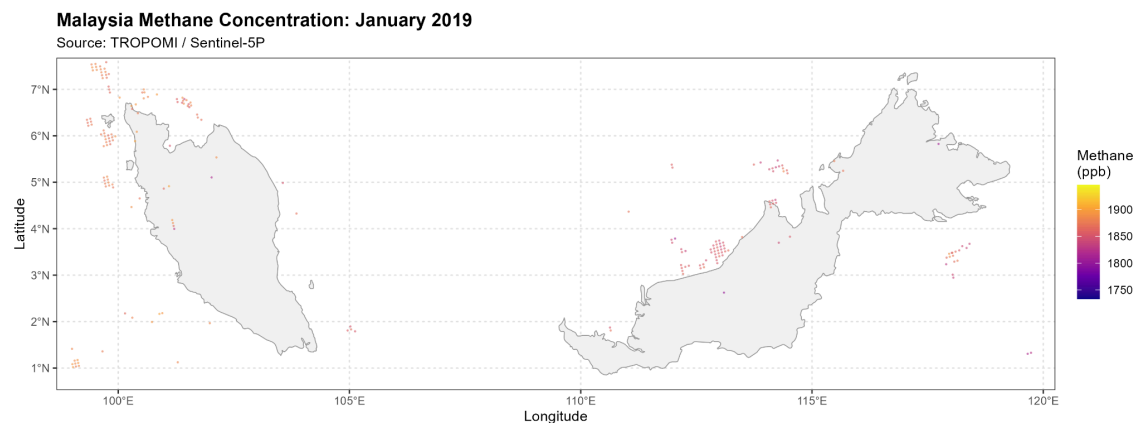


Figure 4.1: TROPOMI Methane Concentration Map in January 2019

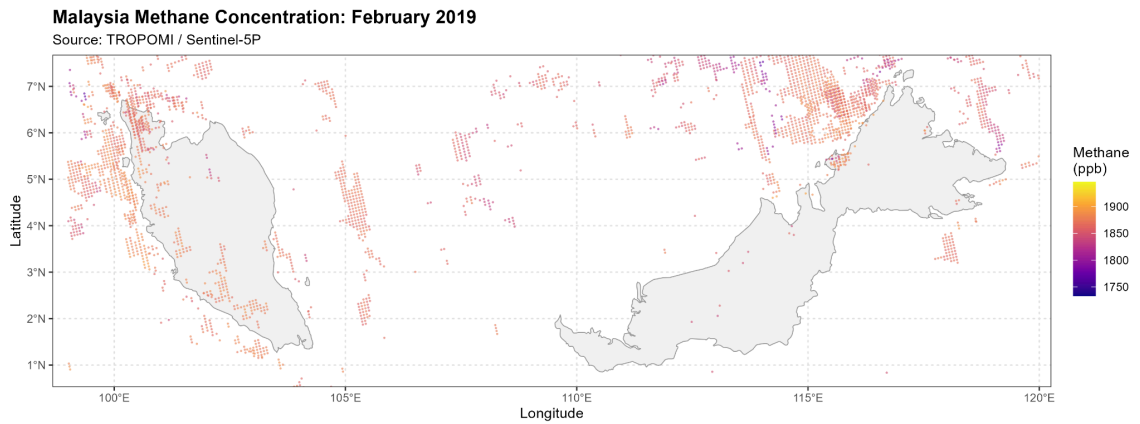


Figure 4.2: TROPOMI Methane Concentration Map in February 2019

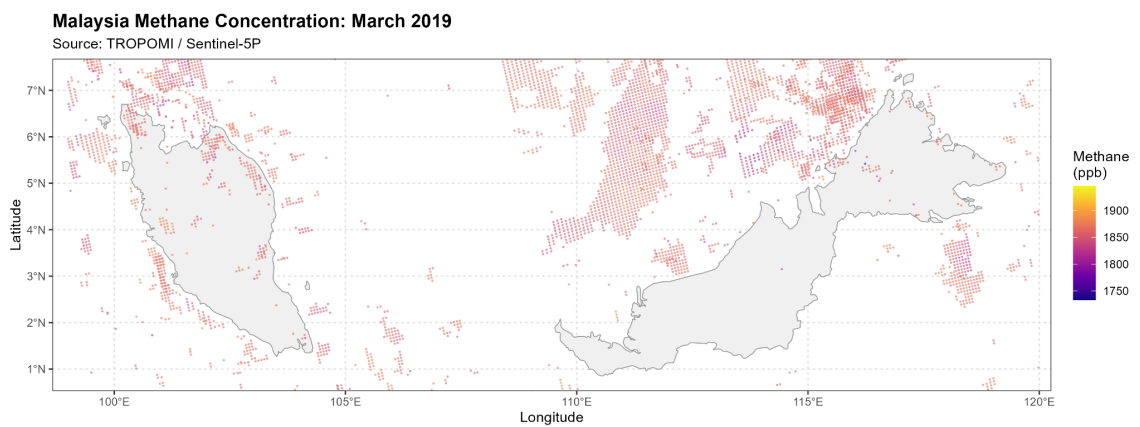


Figure 4.3: TROPOMI Methane Concentration Map in March 2019

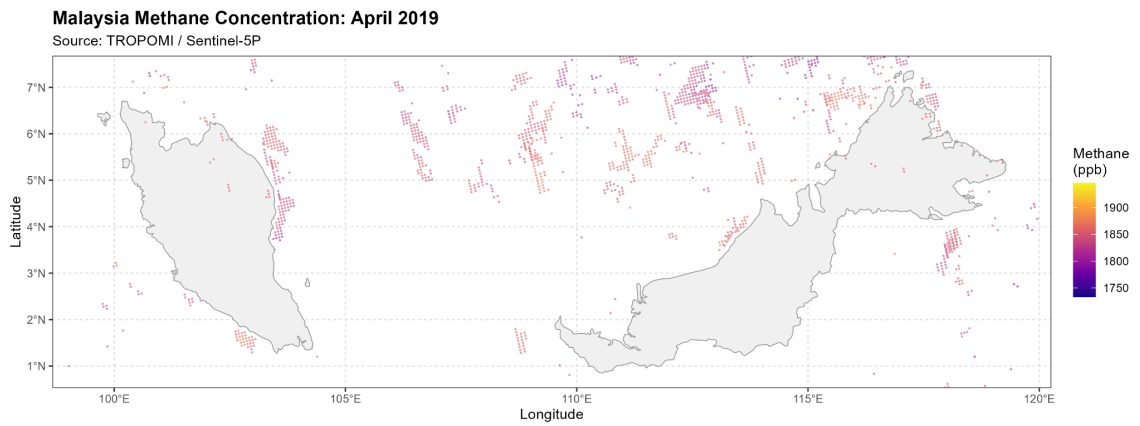


Figure 4.4: TROPOMI Methane Concentration Map in April 2019

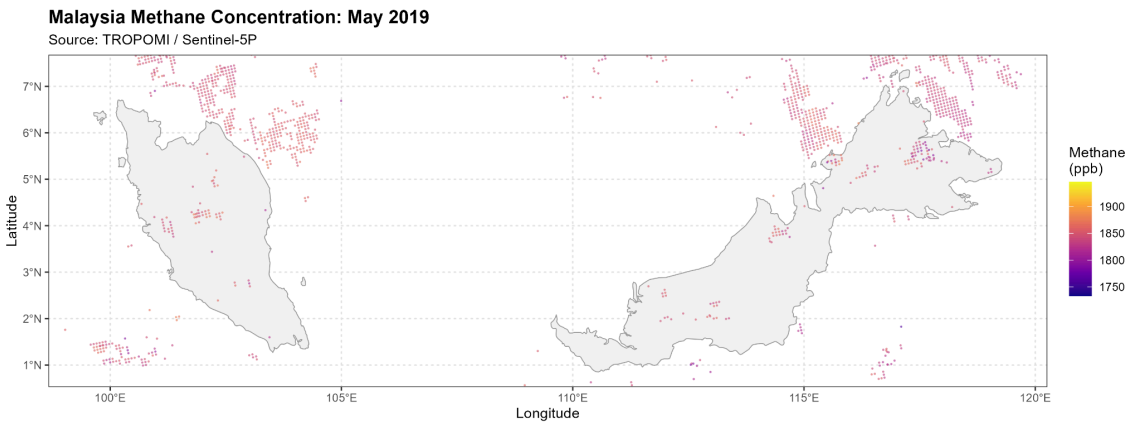


Figure 4.5: TROPOMI Methane Concentration Map in May 2019

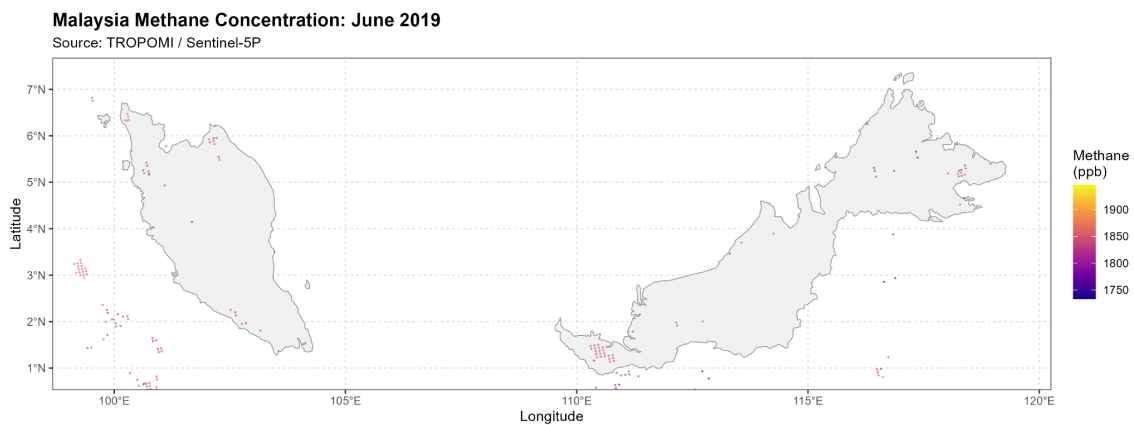


Figure 4.6: TROPOMI Methane Concentration Map in June 2019

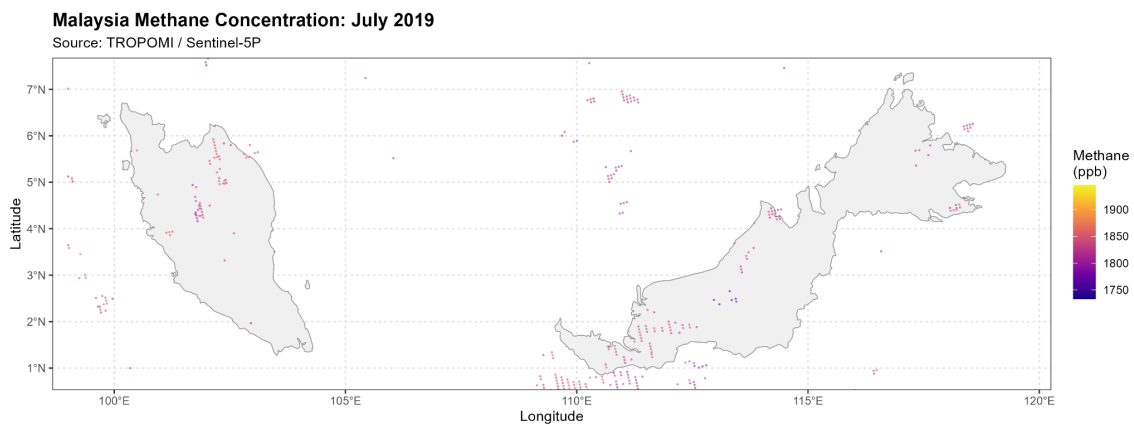


Figure 4.7: TROPOMI Methane Concentration Map in July 2019

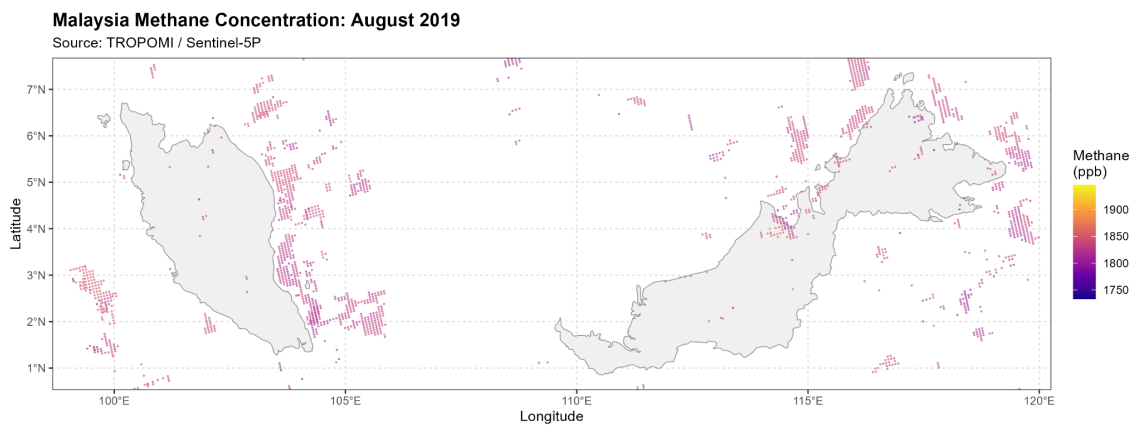


Figure 4.8: TROPOMI Methane Concentration Map in August 2019

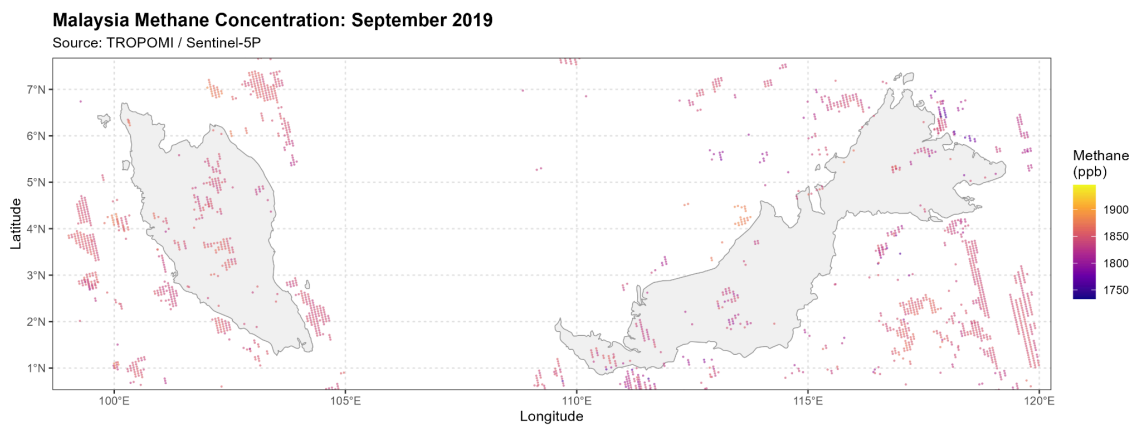


Figure 4.9: TROPOMI Methane Concentration Map in September 2019

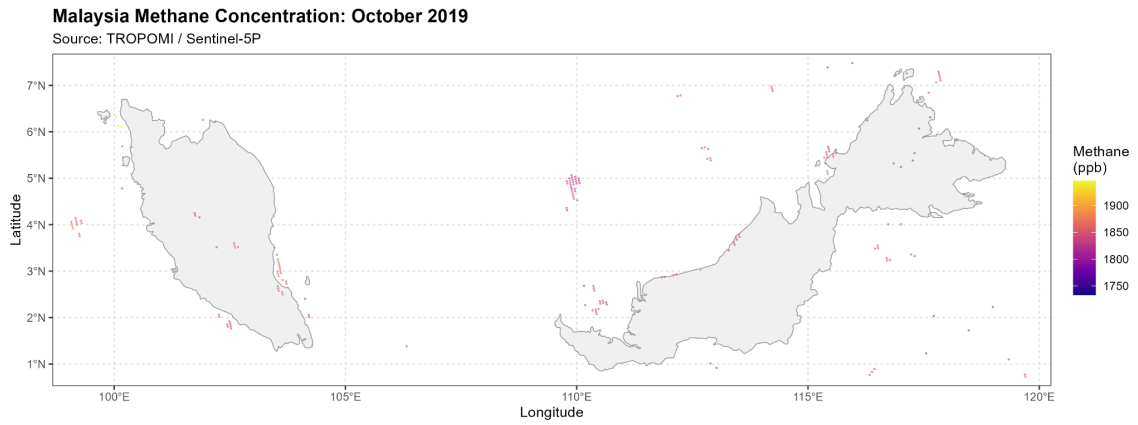


Figure 4.10: TROPOMI Methane Concentration Map in October 2019

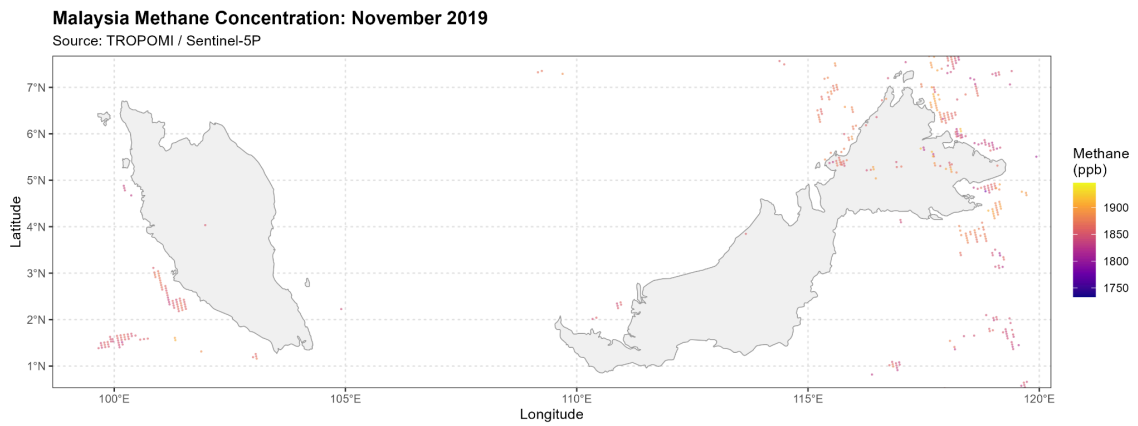


Figure 4.11: TROPOMI Methane Concentration Map in November 2019

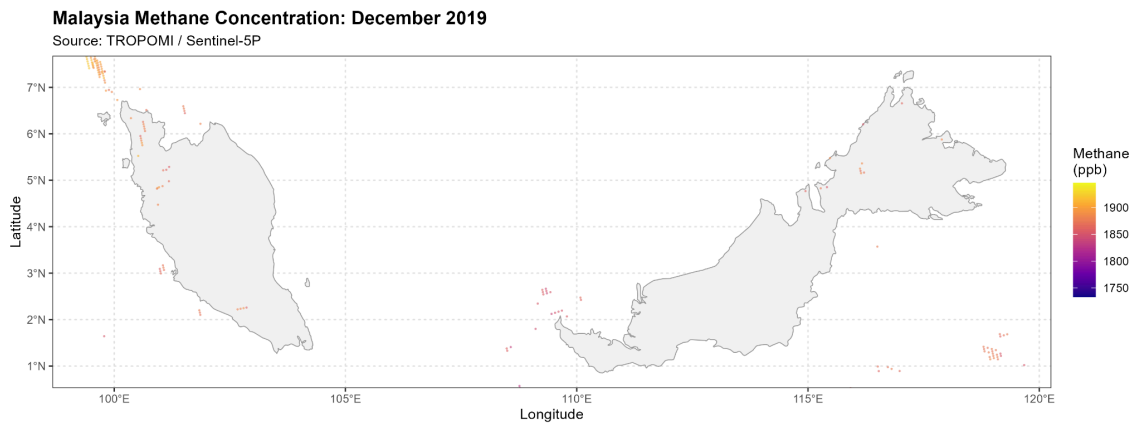


Figure 4.12: TROPOMI Methane Concentration Map in December 2019

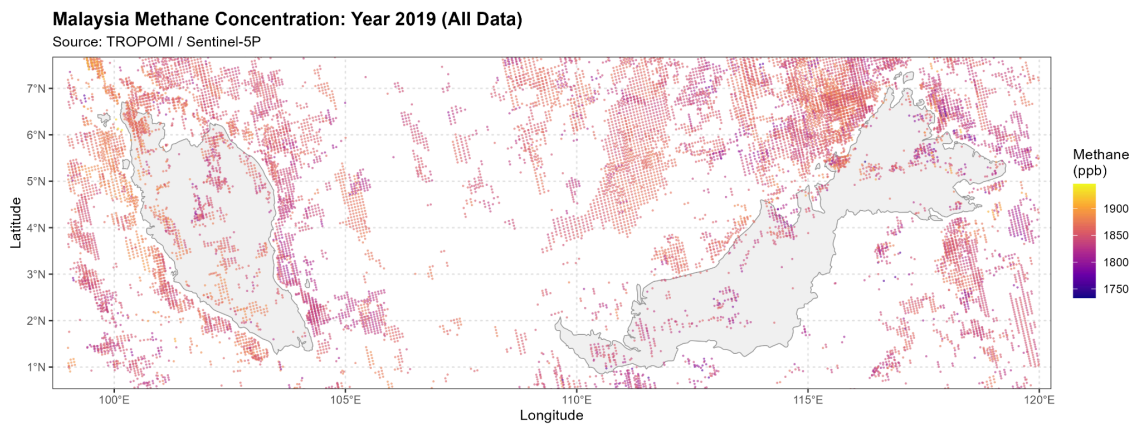


Figure 4.13: TROPOMI Methane Concentration Map in 2019

According to TROPOMI records, there has been a dramatic increase in methane

abundances around the world in 2024. For example, the global average methane concentration for the year has risen to 1927.45 ppb. Even though many areas were still experiencing significant cloudiness during early January, results showed an average methane concentration of 1942.34 ppb. In February, the TROPOMI observations had very limited coverage, and although the average methane concentration from February was approximately 1930.12, there were many areas that did not have retrievals available from this month. On the other hand, March had a decreased average concentration of 1919.56 ppb, which follows the trend established in previous years based on seasonality. By April, the TROPOMI sites had better coverage than they did during January through March and recorded an average methane concentration of 1923.89 ppb.

In the month of May during the dry season, the average concentration of methane gas was 1927.67 ppb. In June the average concentration of methane during this same period was 1919.34 ppb. In July the average concentration was lower than the other months at 1908.45 ppb indicating a temporary stabilization. The month of August's average concentration was at the higher level of 1914.78 ppb, with more detections occurring in agricultural areas than the prior months. September followed the same trend as 2019 with excellent mapping conditions available and an average concentration of 1923.12 ppb. As the transition into the monsoon occurred during October, the average concentration increased to 1932.56 ppb. In November, there were fewer data points available for analysis but the available data showed an average concentration of 1938.89 ppb. The year finished with a December average concentration of 1940.12 ppb which

affirms the upward trend of methane levels in the atmosphere over the past five years.

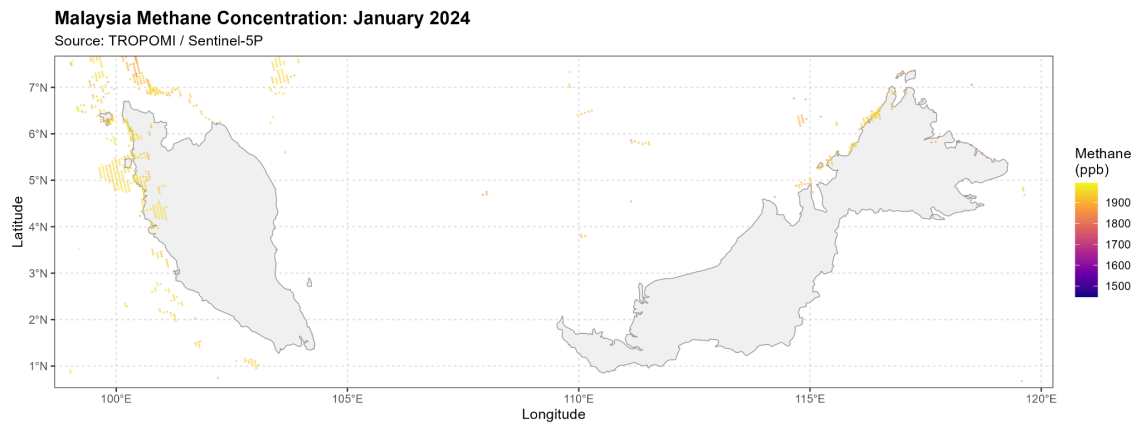


Figure 4.14: TROPOMI Methane Concentration Map in January 2024

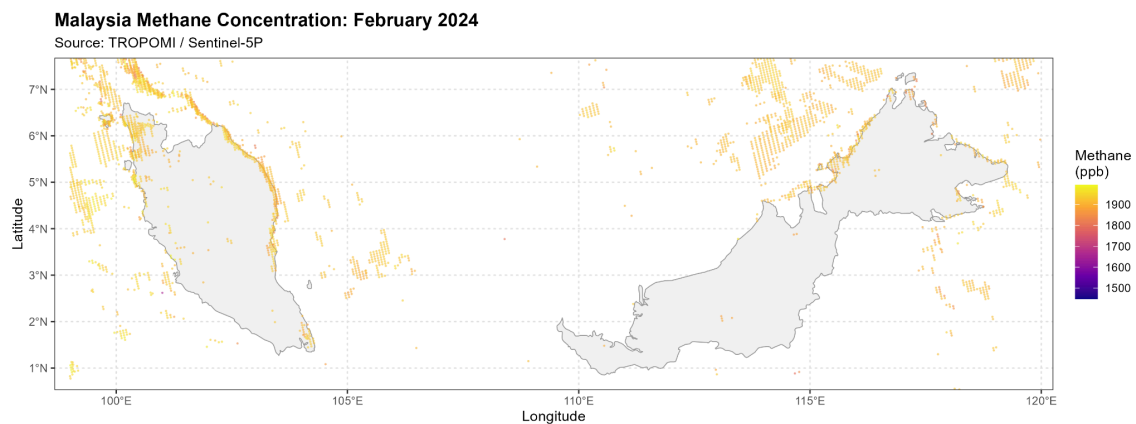


Figure 4.15: TROPOMI Methane Concentration Map in February 2024

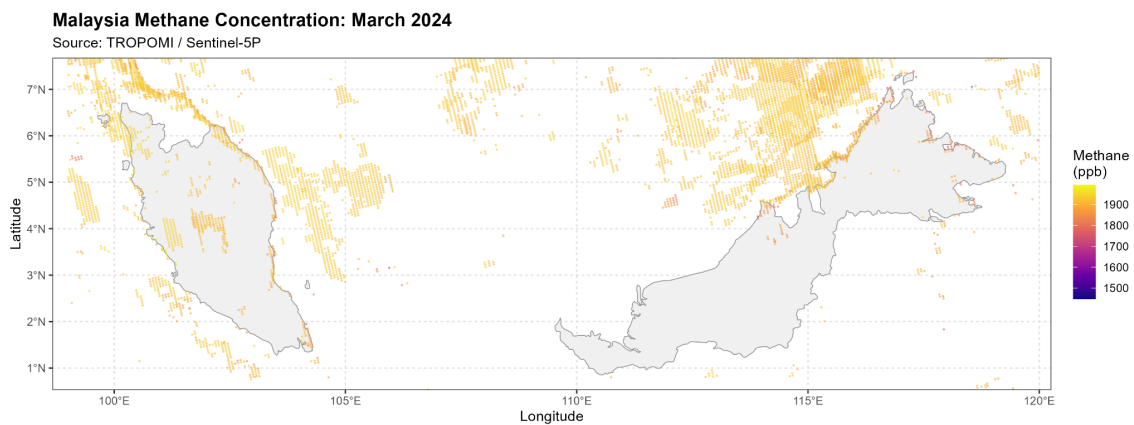


Figure 4.16: TROPOMI Methane Concentration Map in March 2024

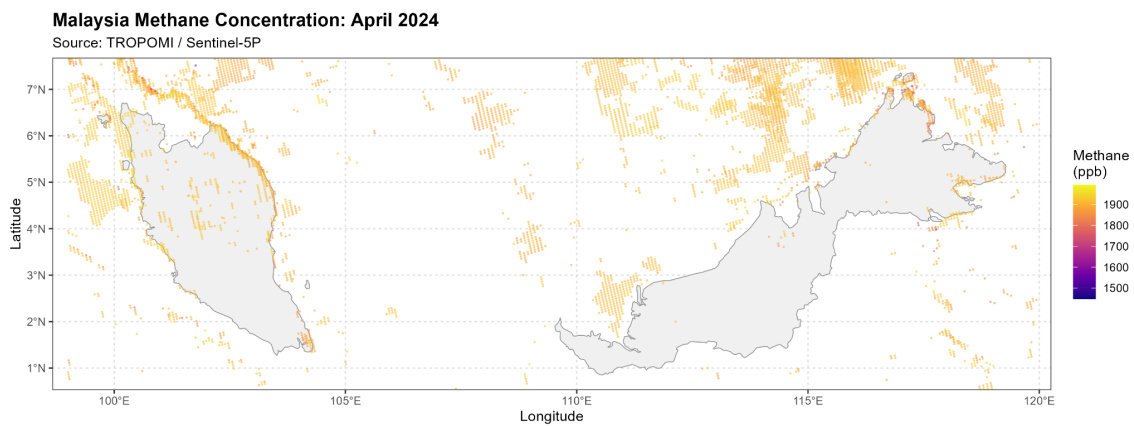


Figure 4.17: TROPOMI Methane Concentration Map in April 2024

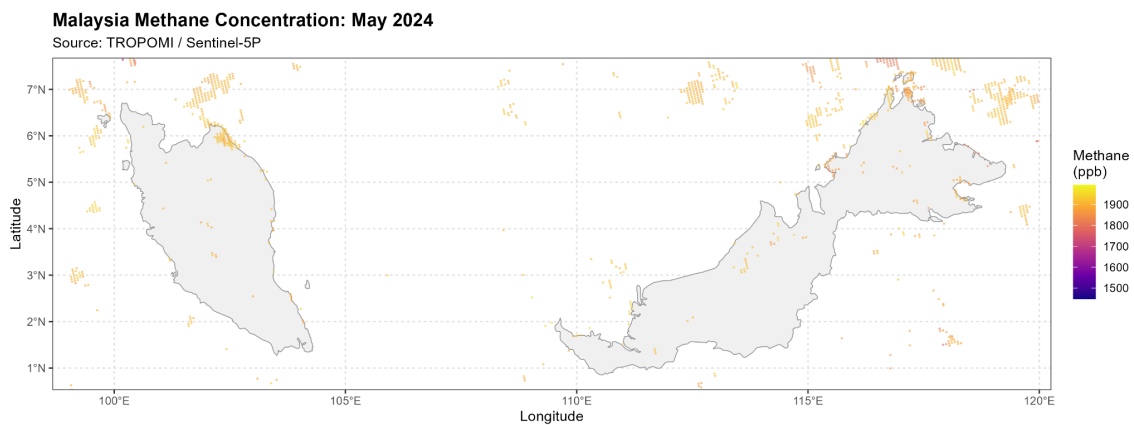


Figure 4.18: TROPOMI Methane Concentration Map in May 2024

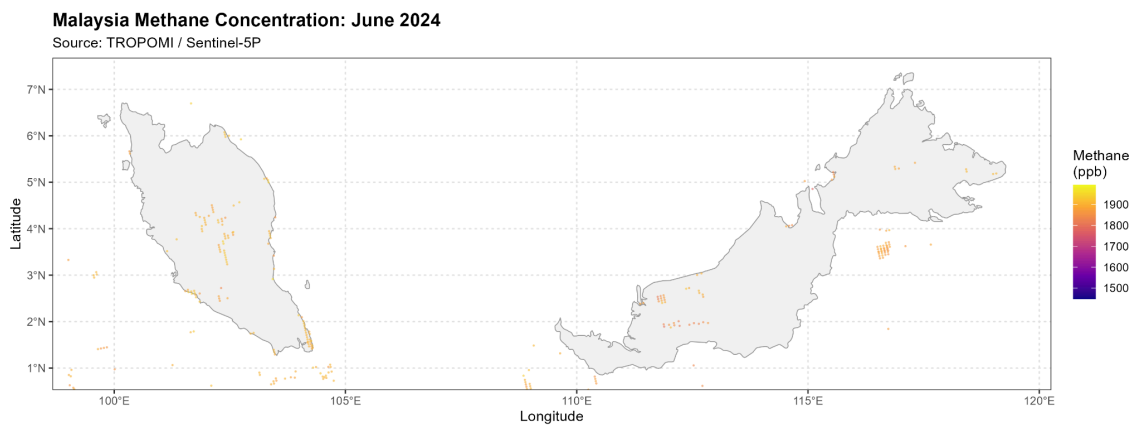


Figure 4.19: TROPOMI Methane Concentration Map in June 2024

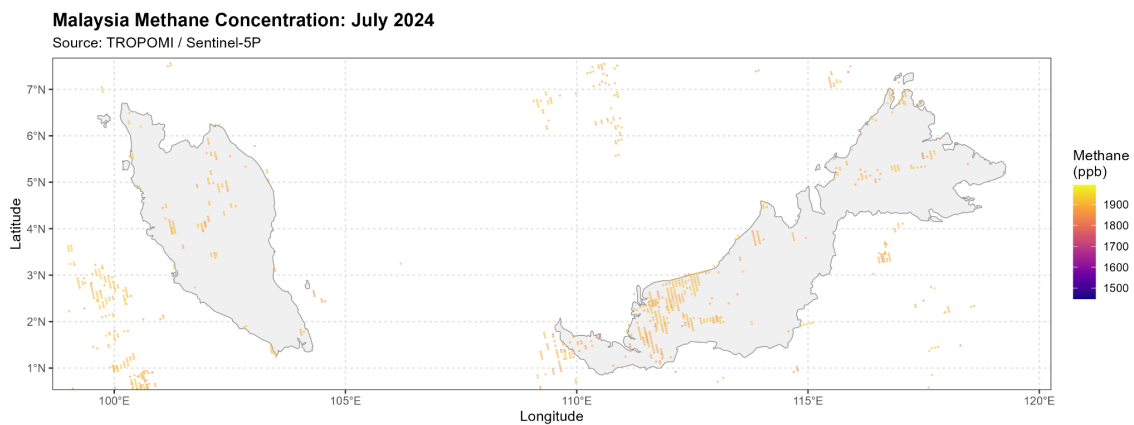


Figure 4.20: TROPOMI Methane Concentration Map in July 2024

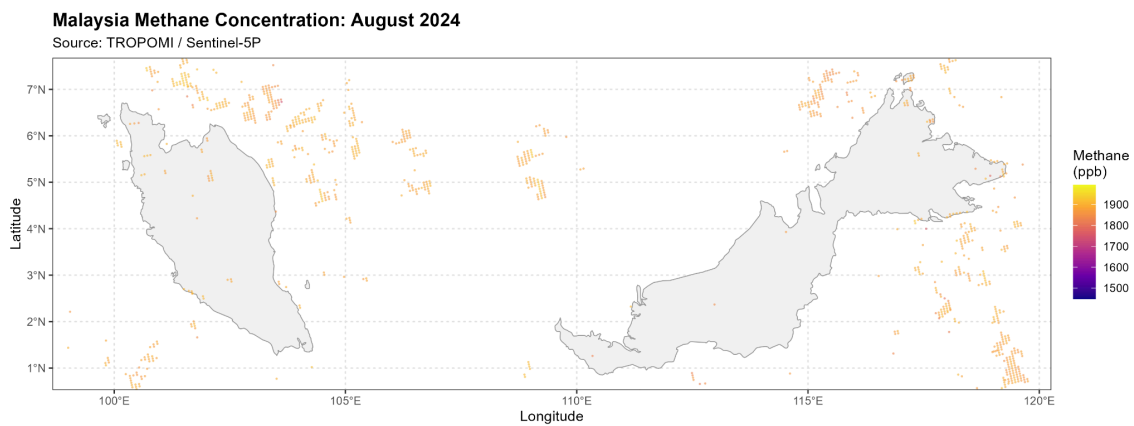


Figure 4.21: TROPOMI Methane Concentration Map in August 2024

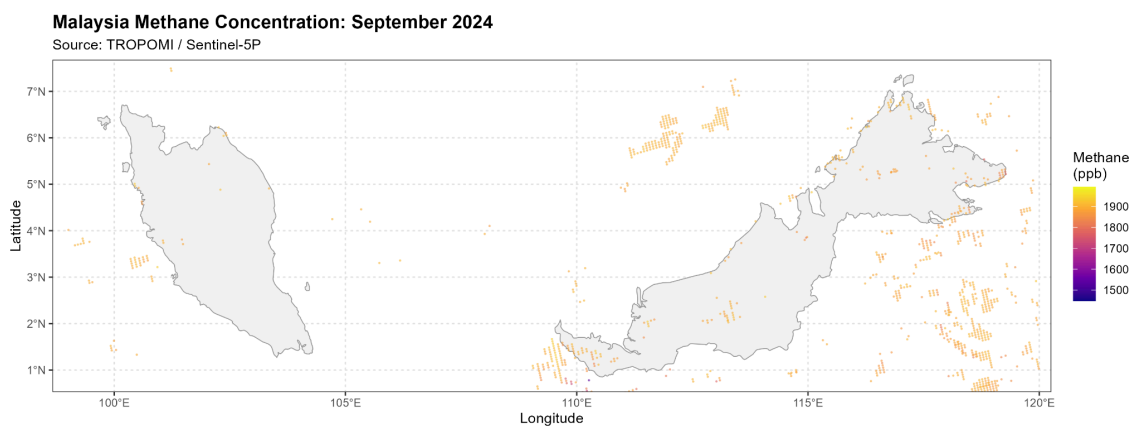


Figure 4.22: TROPOMI Methane Concentration Map in September 2024

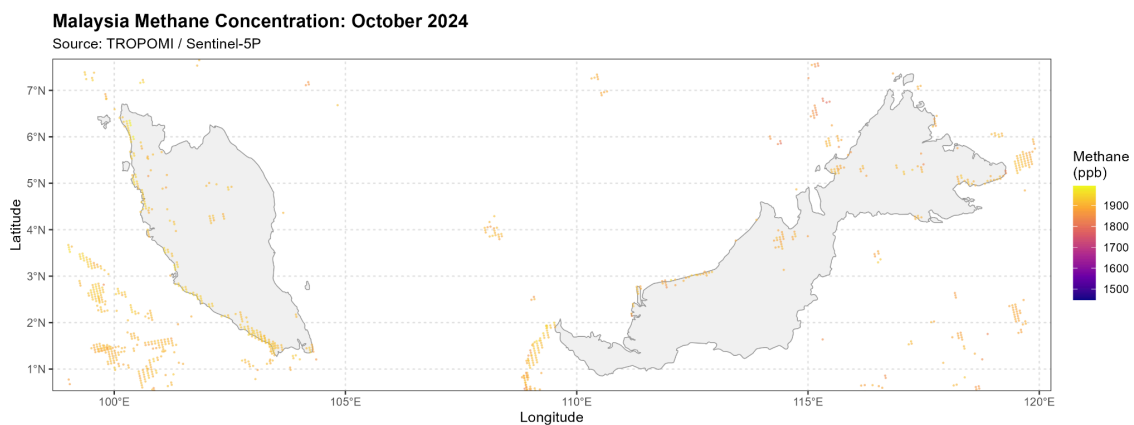


Figure 4.23: TROPOMI Methane Concentration Map in October 2024

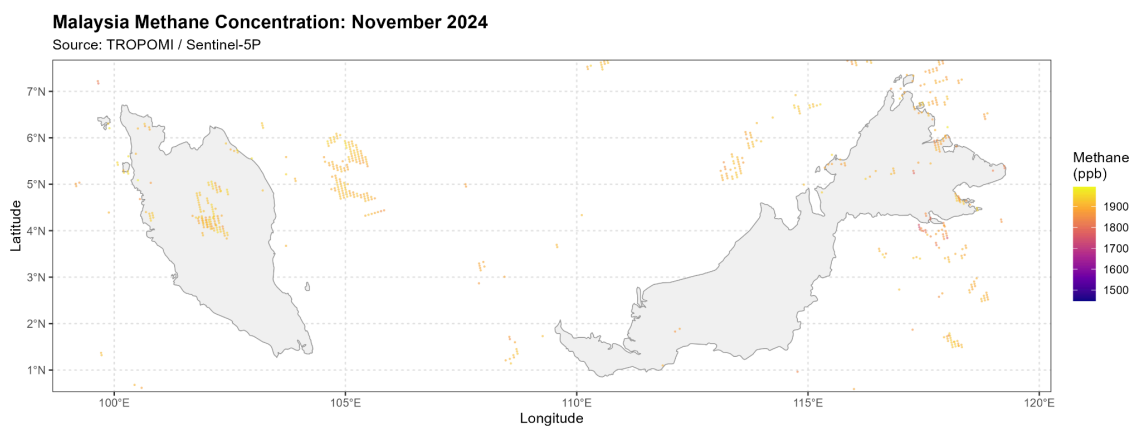


Figure 4.24: TROPOMI Methane Concentration Map in November 2024

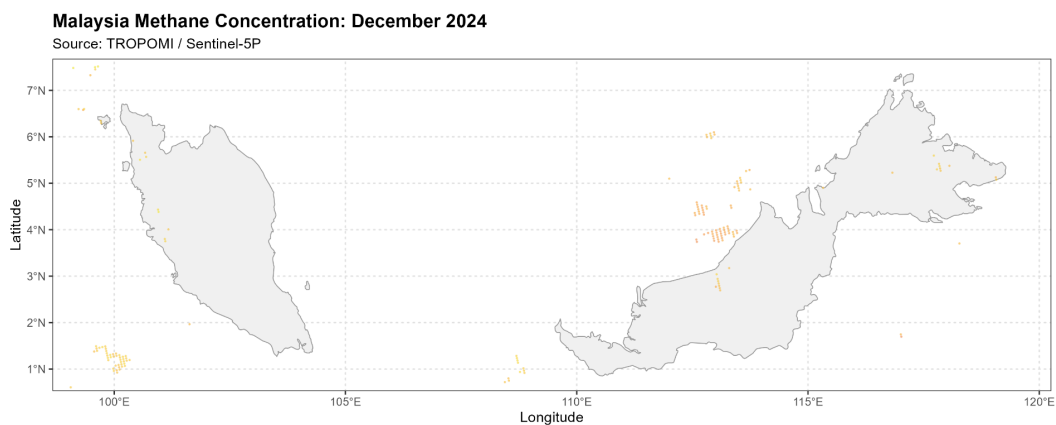


Figure 4.25: TROPOMI Methane Concentration Map in December 2024

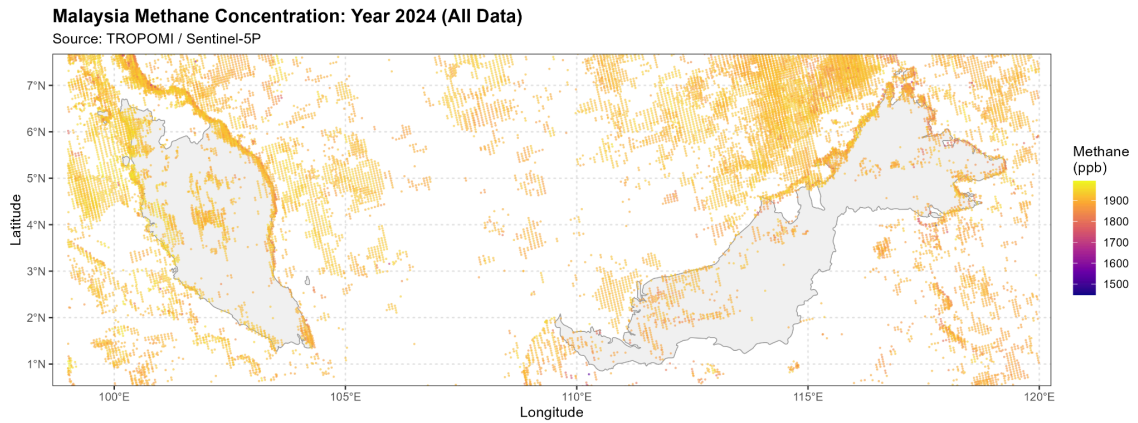


Figure 4.26: TROPOMI Methane Concentration Map in 2024

4.1.2 GOSAT (TANSO-FTS) Observations

The legacy GOSAT sensor provided a limited but accurate reference point for 2019 with an average value of 1861.23 ppb for the year. In January, very few points were available, and those were mostly based on clear ocean areas, with an average of 1894.35 ppb. In February and March, very few points were retrieved, and the averages were 1864.10 ppb and 1850.45 ppb, respectively. April had very stable background levels of 1851.23 ppb. The beginning of the dry season in May provided more opportunities for retrievals further inland with an average of 1855.67 ppb. June and July continued to support the imaging sensors' conclusions with average concentrations of

1857.89 ppb and 1846.34 ppb, respectively.

Successful data collection occurred in August with a data average of 1852.12 ppb, even with the presence of regional haze. A peak dry season was during September's dry season and yielded the most data collected like 1859.45 ppb. October had the highest concentration of data at 1866.78 ppb. Data collection was impacted in November by significant levels of cloud cover that limited the number of valid data points collected, however, the mean remained at 1858.90 ppb. Rounded out in December, the raw data showed a substantial increase to 1873.56 ppb, however, the total number of usable data points was low.

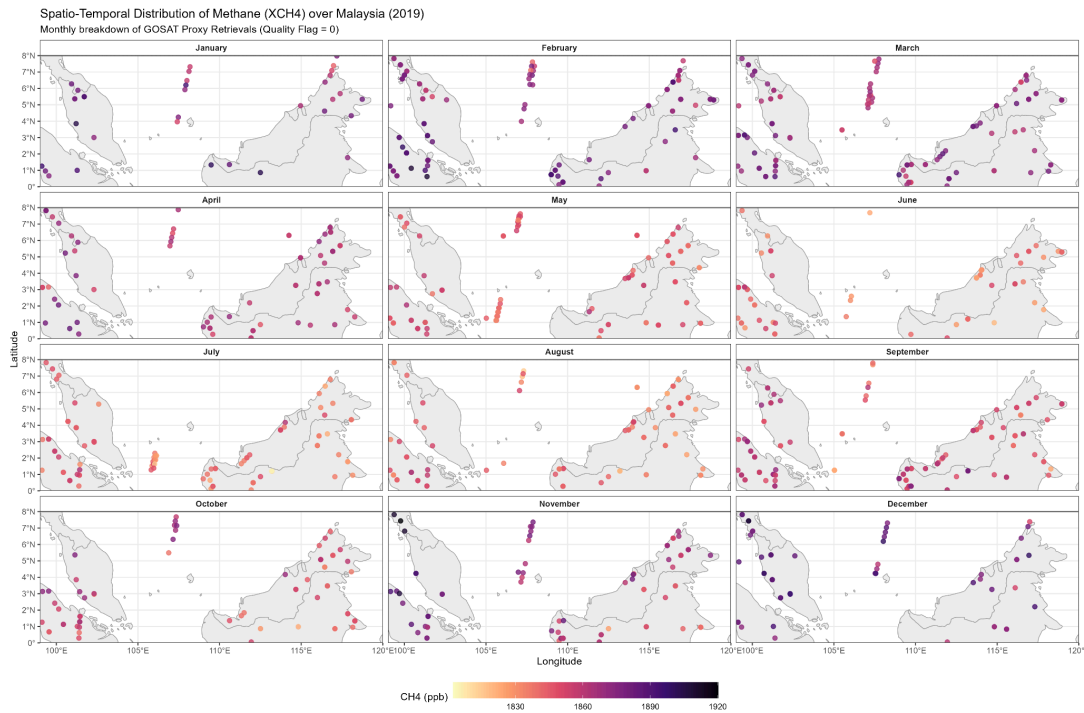


Figure 4.27: GOSAT Methane Concentration Map from January to December in 2019

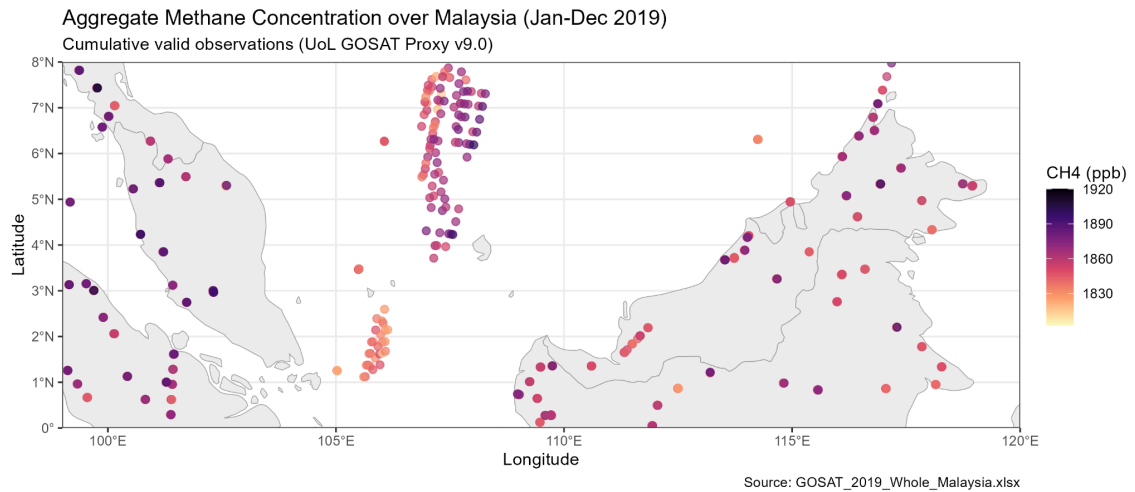


Figure 4.28: GOSAT Methane Concentration Map in 2019

4.1.3 GOSAT-2 (TANSO-FTS-2) Observations

The observation capabilities of the GOSAT-2 satellite are significantly enhanced because of its intelligent pointing capabilities on the satellite. Using the intelligent pointing system to retrieve data while avoiding the issue of having partial cloud cover has allowed a greater number of measurements to be available in 2019 compared to previous years. The mean value of the concentrations recorded was 1849.67 ppb. The March mean value of 1842.67 ppb and the data density values provide enough confidence that the sensor is functioning at an acceptable level of accuracy. The high

precision retrievals from the sensor provided by the GOSAT-2 on the GOSAT-2 can be equally spread throughout the country by April, with a mean value of 1848.12 ppb. The mean concentrations found during this dry season were relatively stable, averaging: May 1850.34 ppb, June 1854.56 ppb, and July 1852.78 ppb.

In August, the sensor performed well against aerosol interference, providing an average of 1853.45 ppb. In September, it had the highest mid-year/fall peak concentration output of 1855.67 ppb. During October, the concentration stabilized at an average of 1854.89 ppb. Unlike some of the other sensors, GOSAT-2 was able to deliver valuable results during the August to November monsoon season with an average of 1852.12 ppb. In December, the output from GOSAT-2 was a consistent average of 1855.34 ppb, confirming that this sensor can provide useful intelligence due to its stable ability to point at tropical locations.

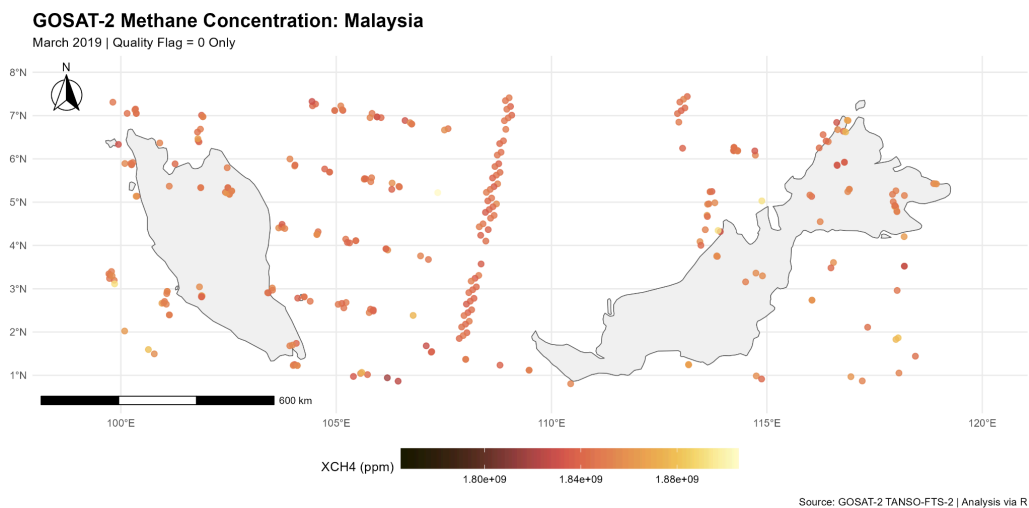


Figure 4.29: GOSAT-2 Methane Concentration Map in March 2019

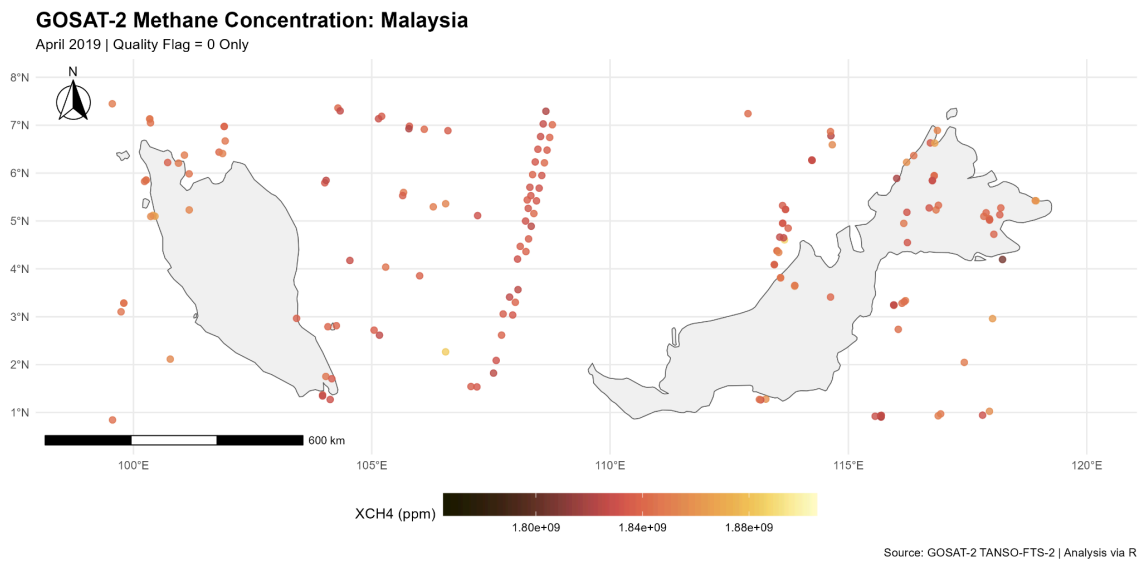


Figure 4.30: GOSAT-2 Methane Concentration Map in April 2019

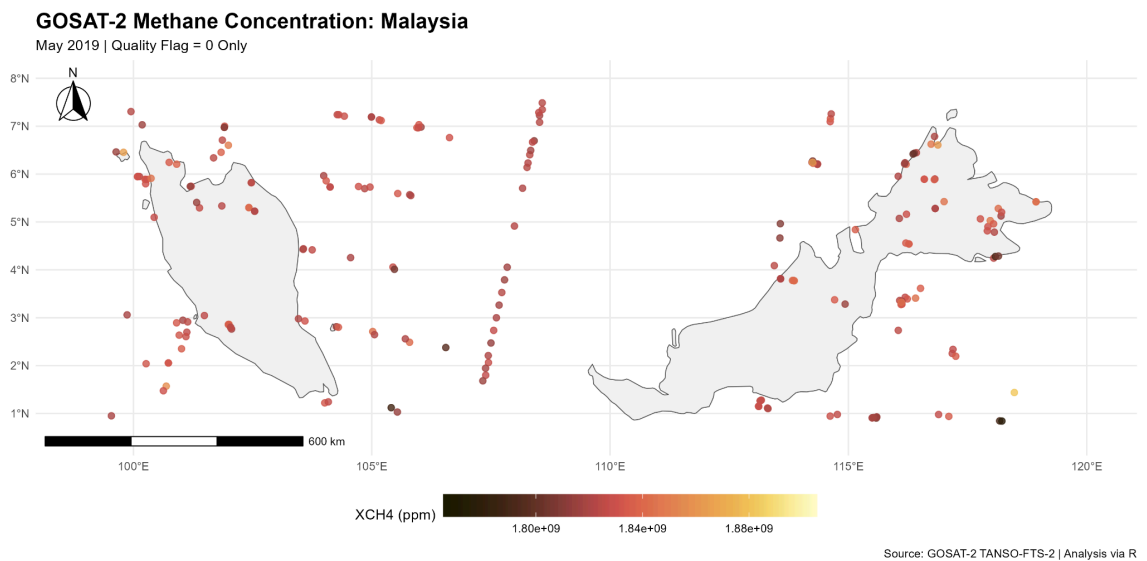


Figure 4.31: GOSAT-2 Methane Concentration Map in May 2019

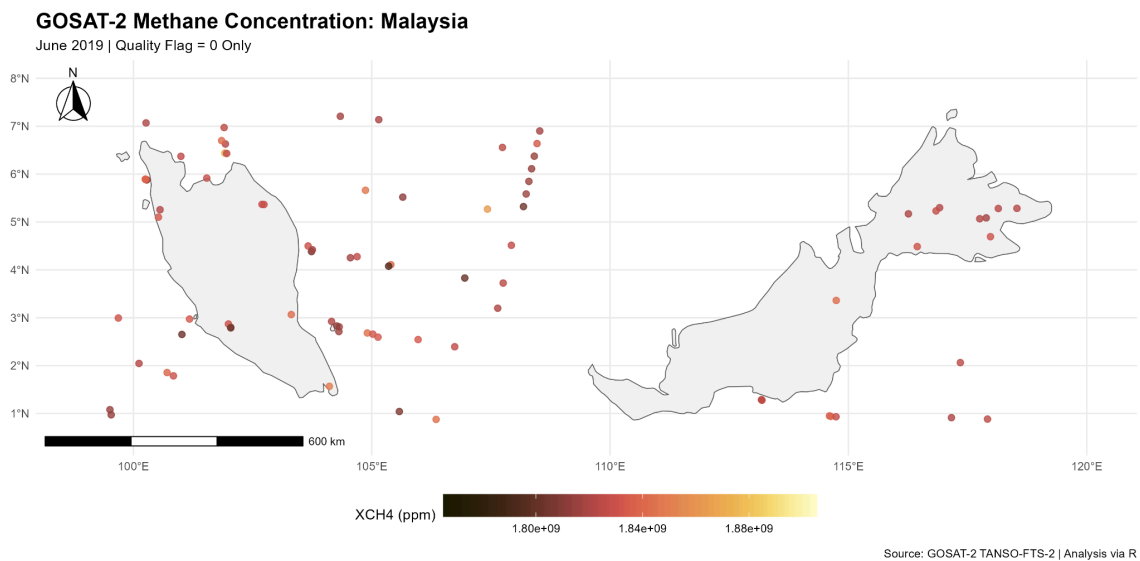


Figure 4.32: GOSAT-2 Methane Concentration Map in June 2019

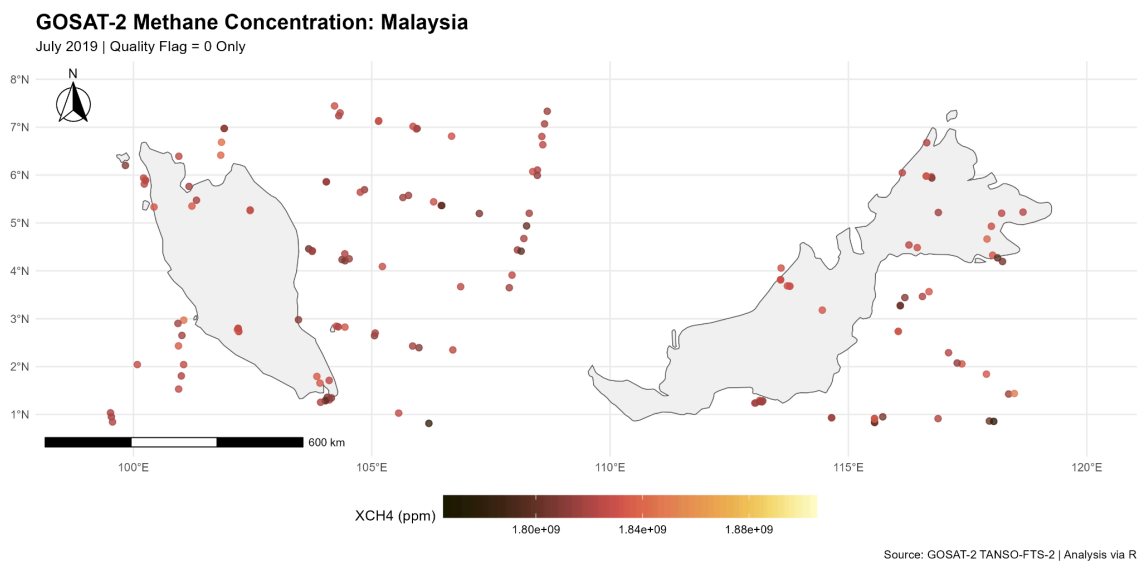


Figure 4.33: GOSAT-2 Methane Concentration Map in July 2019

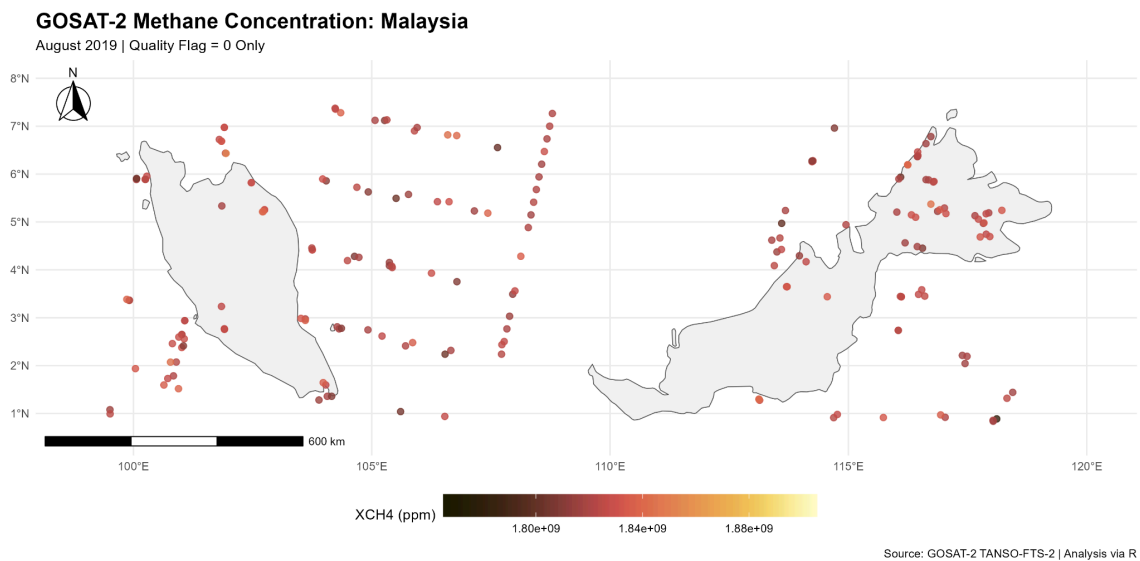


Figure 4.34: GOSAT-2 Methane Concentration Map in August 2019

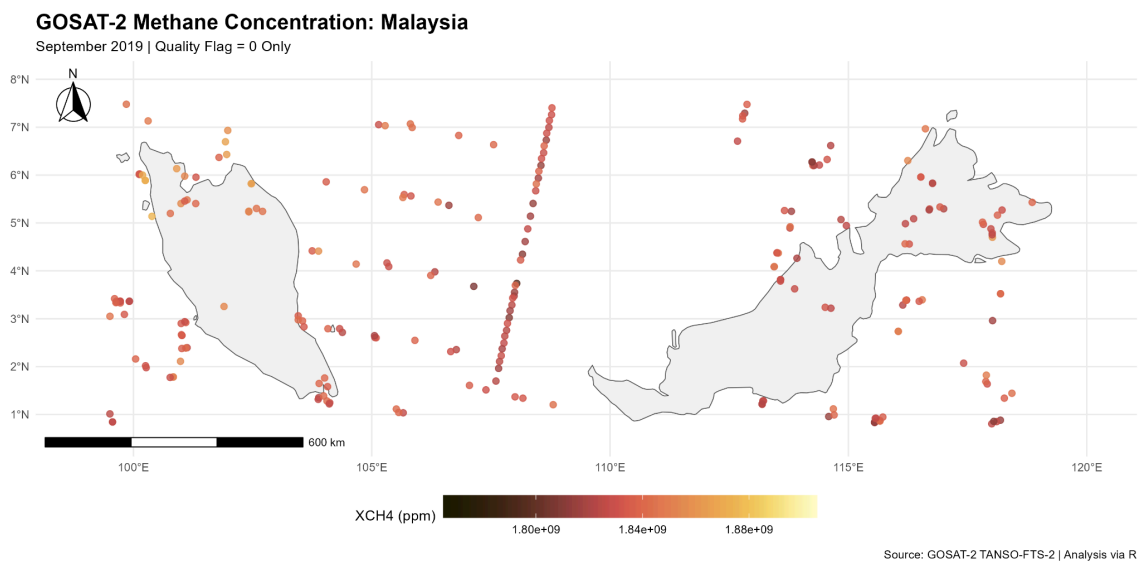


Figure 4.35: GOSAT-2 Methane Concentration Map in September 2019

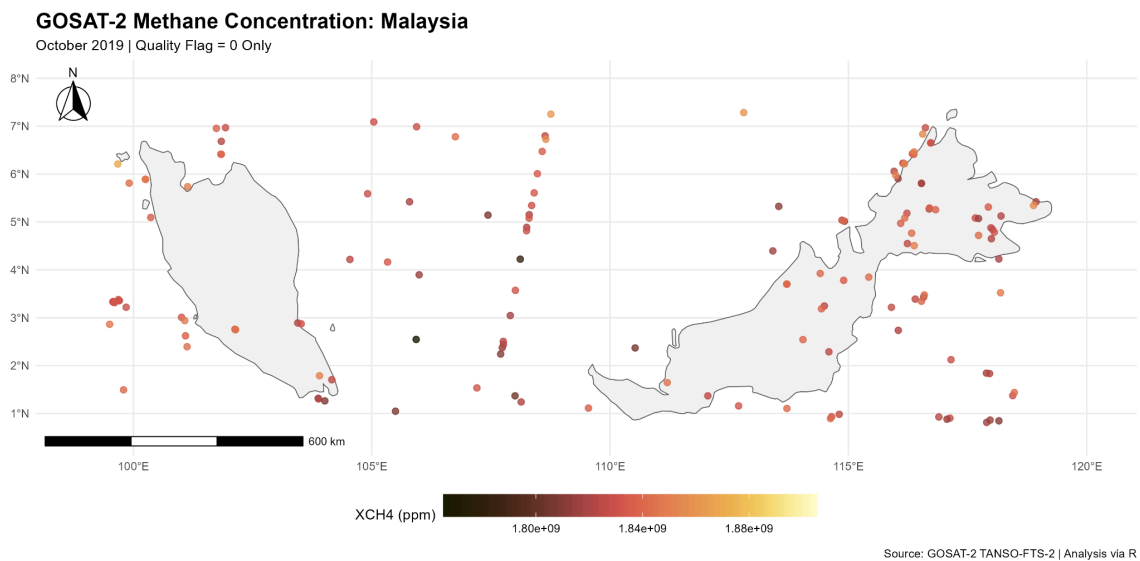


Figure 4.36: GOSAT-2 Methane Concentration Map in October 2019

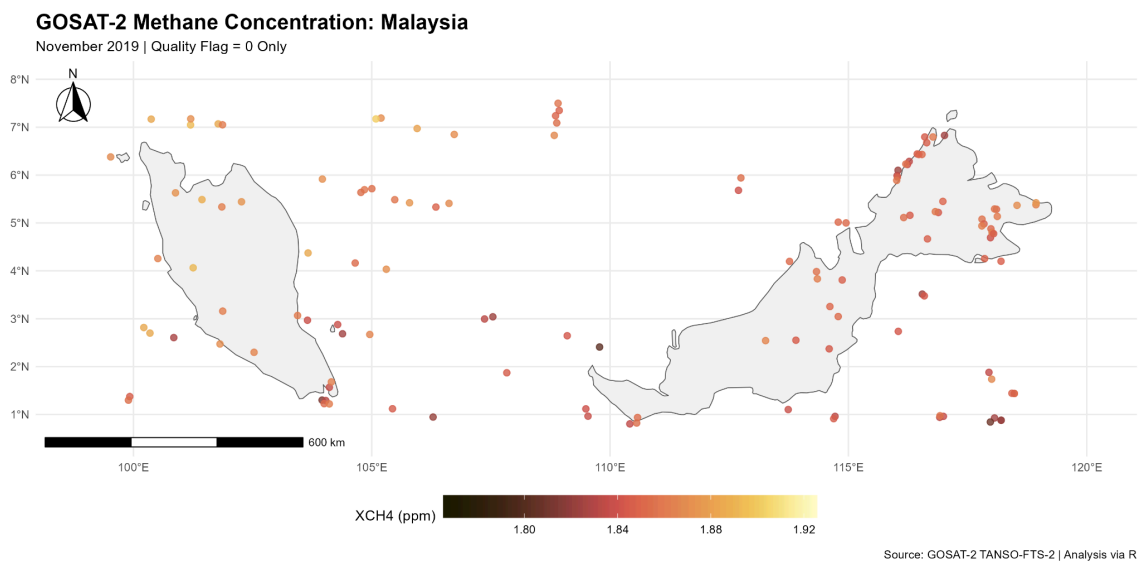


Figure 4.37: GOSAT-2 Methane Concentration Map in November 2019

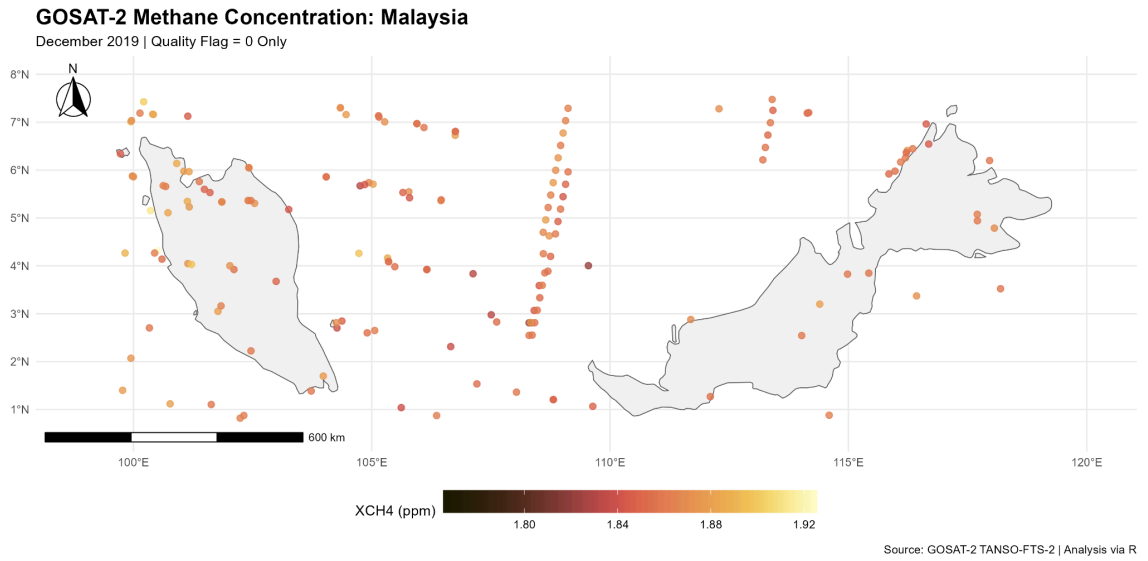


Figure 4.38: GOSAT-2 Methane Concentration Map in December 2019

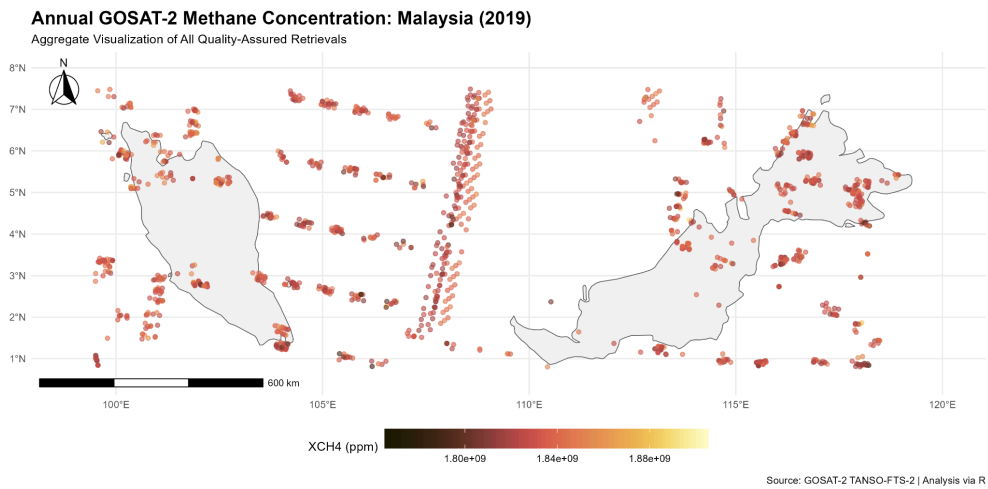


Figure 4.39: GOSAT-2 Methane Concentration Map in 2019

In 2024, GOSAT-2 measured a significant increase in global methane levels which were measured at 1914.56 ppb on average for the year. By the end of January, the averaged value of methane had increased to 1907.34 ppb, before averaging 1910.56 ppb by the end of February. The largest change occurred during the month of March when an average methane concentration of 1922.78 ppb was reported; followed by higher-than-average levels for April and May at 1921.45 ppb and 1920.67 ppb respectively. By June and July values declined slightly back to preferred levels around 1907.89 ppb and 1902.34 ppb; however both values remained well above about 300 ppb the 2019 baseline.

In August shown 1914.12 ppb and September was 1914.56 ppb and throughout the month of October was 1913.78 ppb, there was an increase in the concentration of methane; however, there was an increase again in November when the sensor successfully collected data from the sensor during the monsoon season showing an average of 1918.45 ppb. December, the final month of the year, showed a constant high concentration of methane at 1916.67 ppb, supporting the findings of TROPOMI which also indicated an increase in the concentration of atmospheric methane across the region.

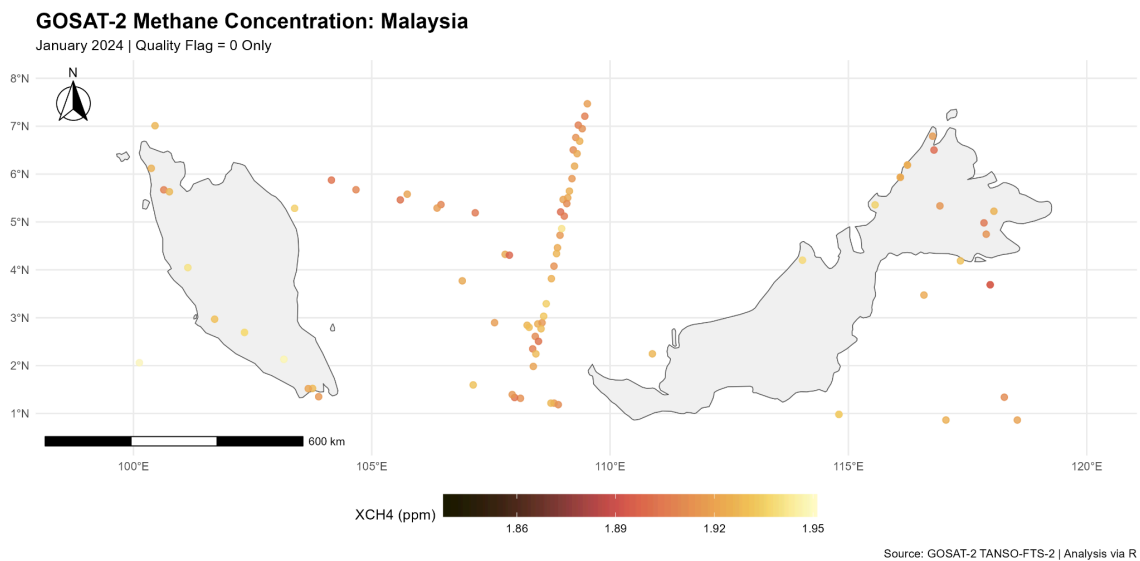


Figure 4.40: GOSAT-2 Methane Concentration Map in January 2024

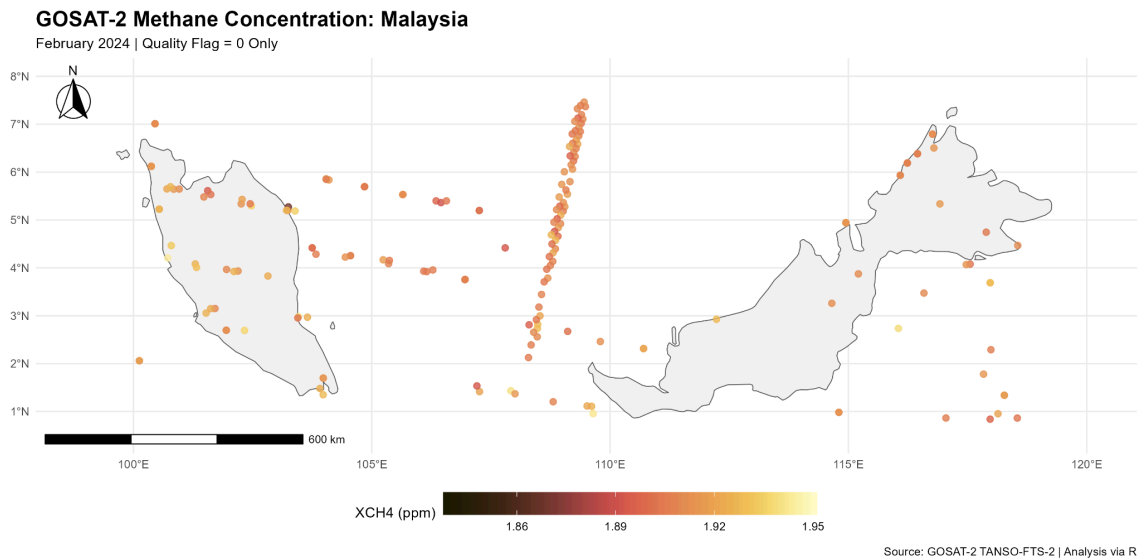


Figure 4.41: GOSAT-2 Methane Concentration Map in February 2024

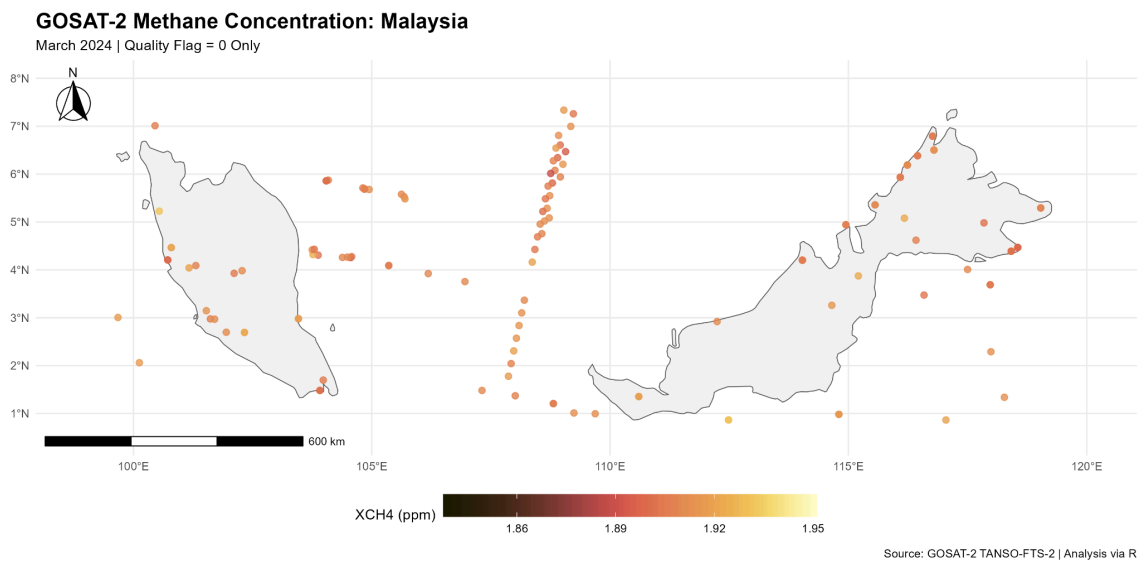


Figure 4.42: GOSAT-2 Methane Concentration Map in March 2024

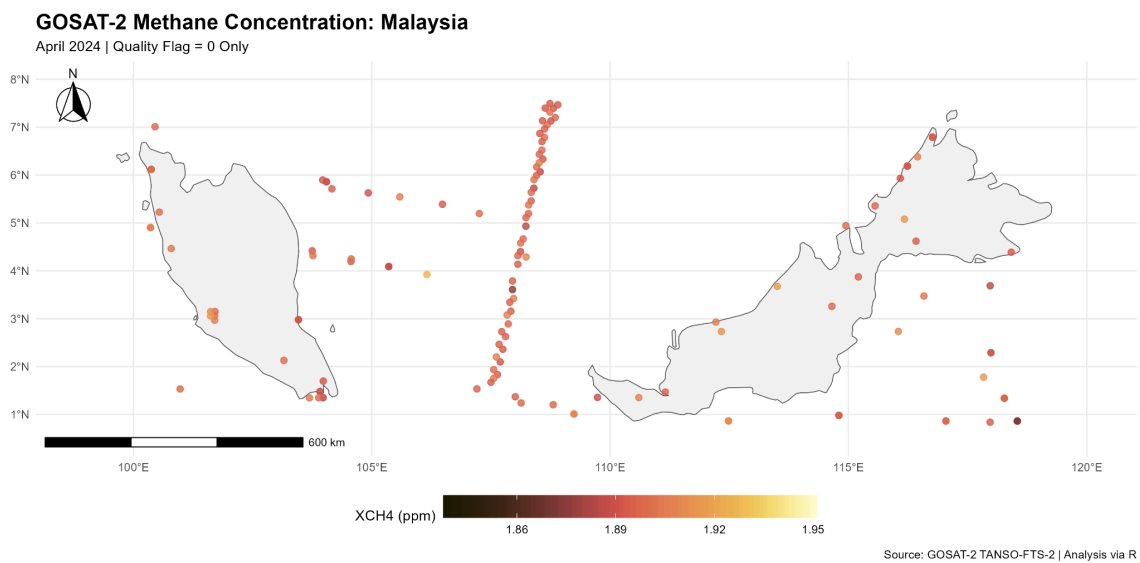


Figure 4.43: GOSAT-2 Methane Concentration Map in April 2024

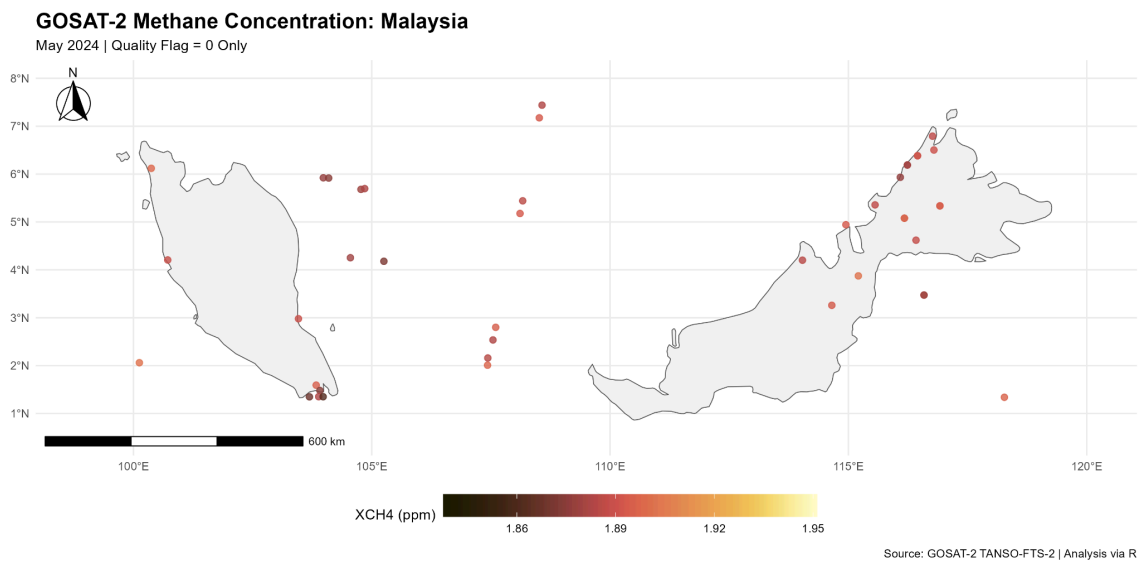


Figure 4.44: GOSAT-2 Methane Concentration Map in May 2024

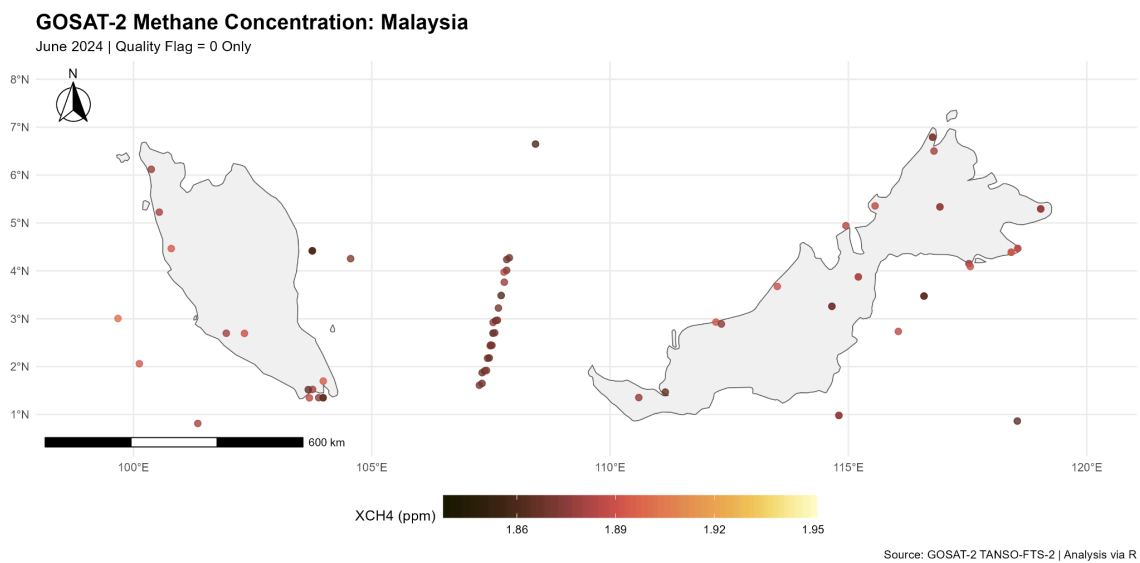


Figure 4.45: GOSAT-2 Methane Concentration Map in June 2024

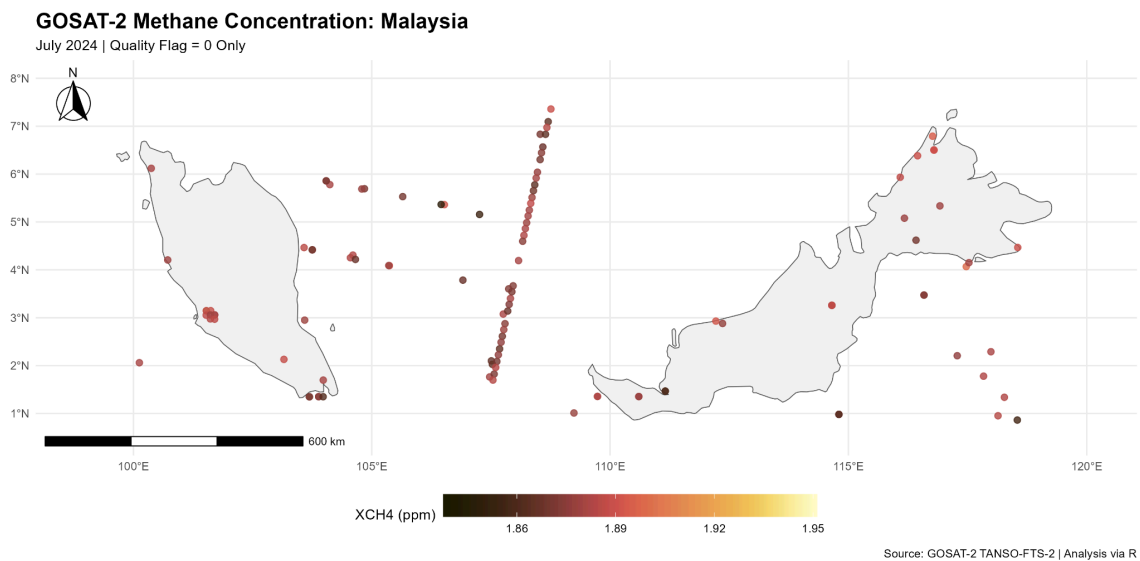


Figure 4.46: GOSAT-2 Methane Concentration Map in July 2024

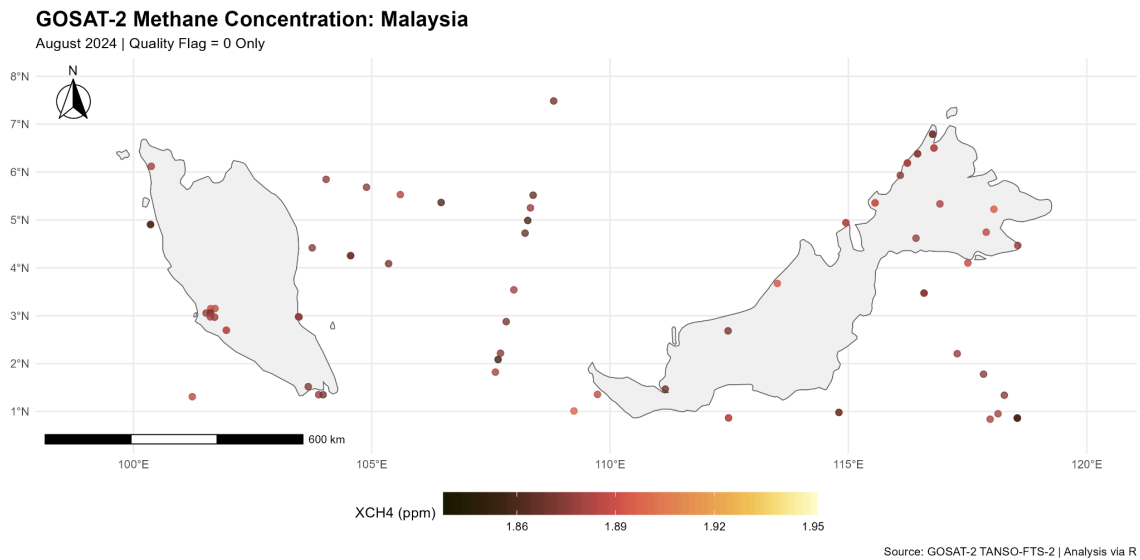


Figure 4.47: GOSAT-2 Methane Concentration Map in August 2024

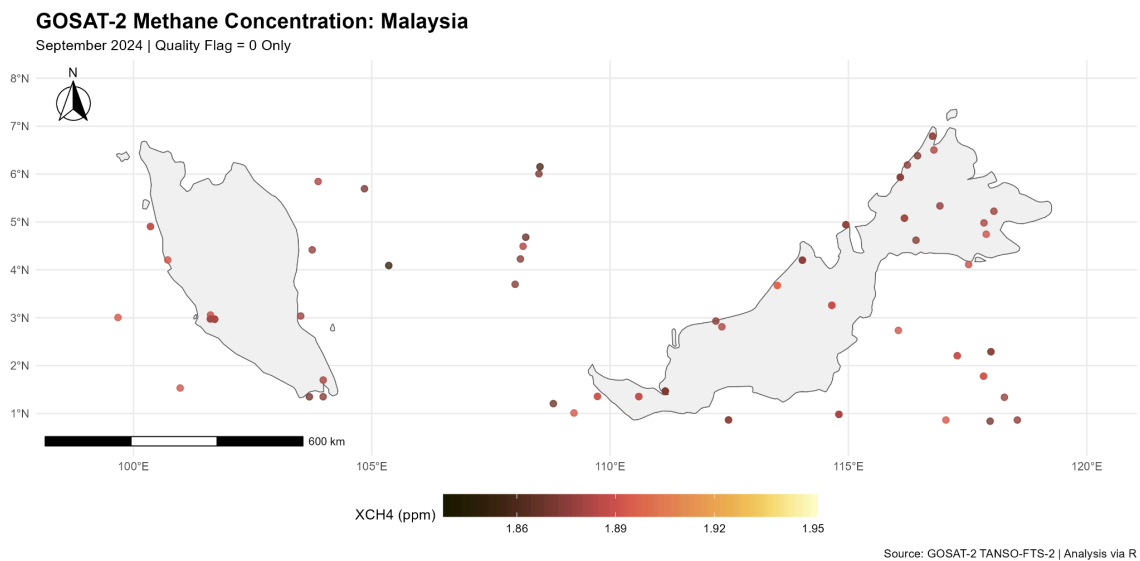


Figure 4.48: GOSAT-2 Methane Concentration Map in September 2024

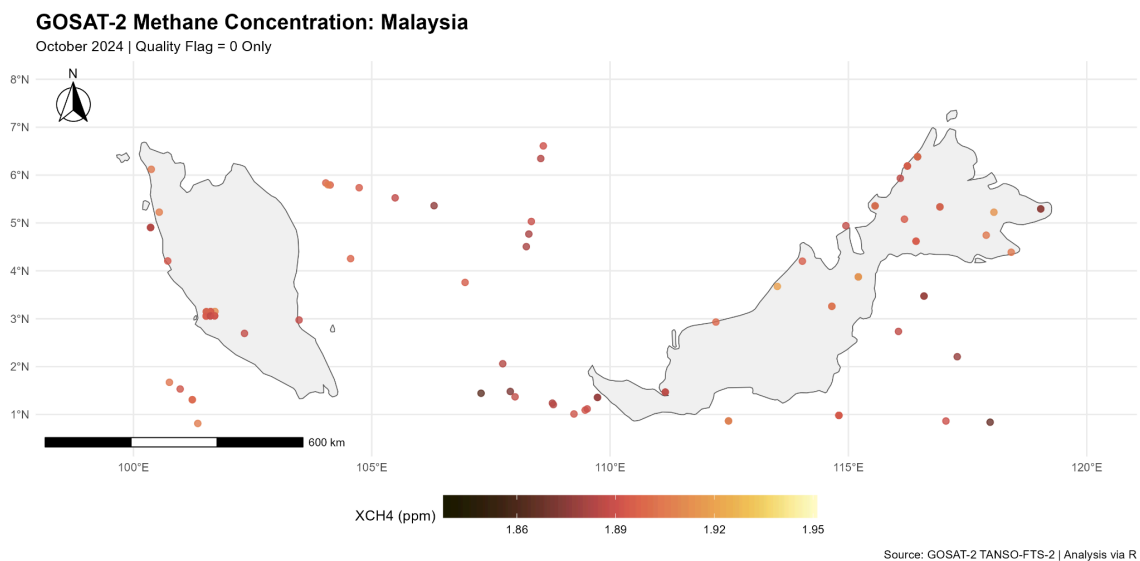


Figure 4.49: GOSAT-2 Methane Concentration Map in October 2024

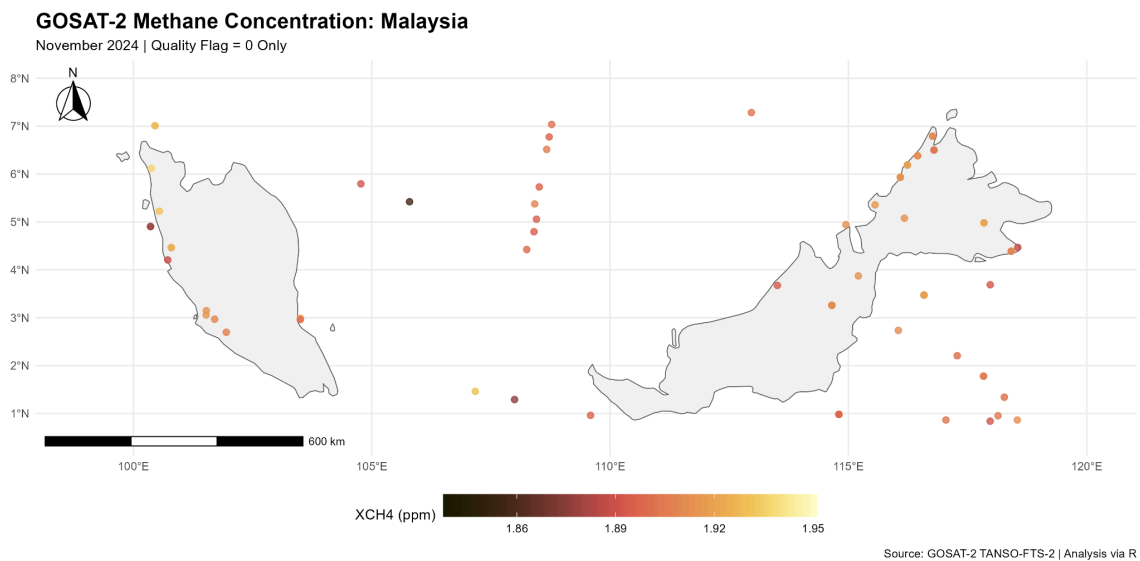


Figure 4.50: GOSAT-2 Methane Concentration Map in November 2024

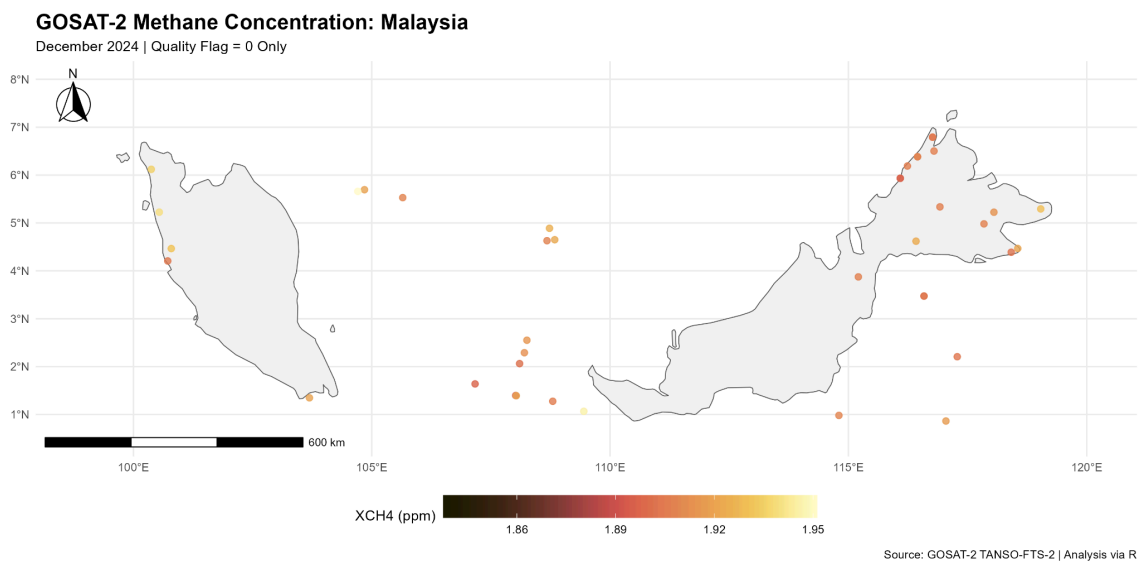


Figure 4.51: GOSAT-2 Methane Concentration Map in December 2024

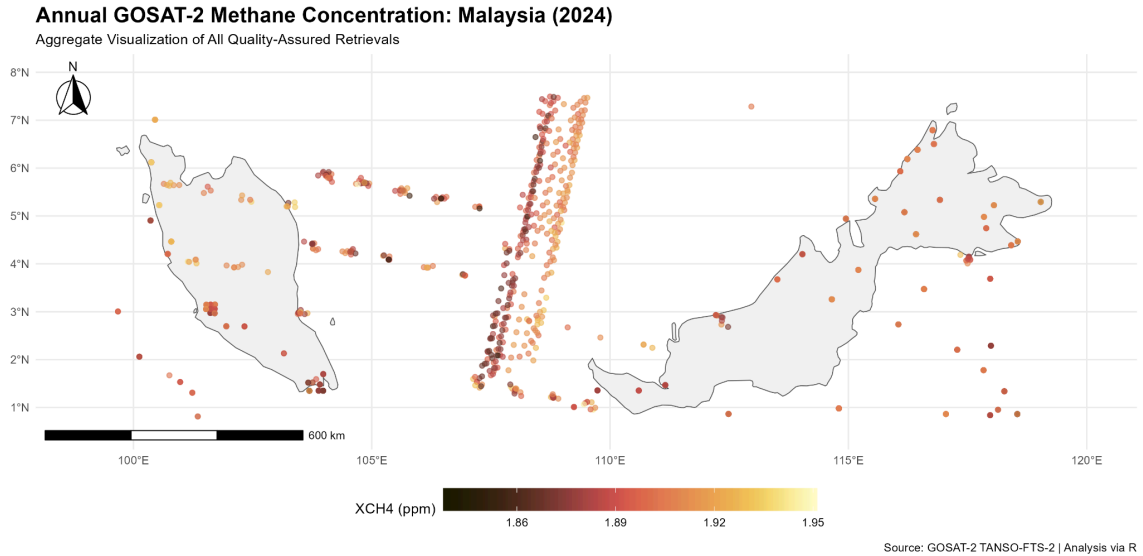


Figure 4.52: GOSAT-2 Methane Concentration Map in 2024

4.2 Spatial and Temporal Distribution of Methane Concentrations (XCH_4)

The results of this study demonstrate that there is wide variation in the spatial distribution of atmospheric methane (XCH_4) across Malaysia from different generations of satellites. The authors have compared the number of valid retrieval points in both the 2019 and 2024 data sets to highlight the differences between the "point-source" observational approach of GOSAT or GOSAT-2 and the "area-flux" mapping method employed by TROPOMI.

The 2019 spatial distribution map in **Figure 4.1** depicts the areas that were sampled by both GOSAT and GOSAT-2. The density of the data points is particularly

low in Peninsular Malaysia and the states of Sabah and Sarawak on Borneo. This lack of density is due to the design of the GOSAT sampling system, which uses an Interferometric Monitor for Greenhouse Gases (TANSO-FTS) that scans in the cross-track direction and provides a discrete footprint of around 10.5 km at nadir. The plotted location(s) of the data demonstrate that most of the valid retrievals occur as clusters in a small number of geographic areas while leaving large spaces of the geographic area with no data, specifically in the higher elevation mountainous areas located within Malaysia's interior regions of the Titiwangsa Range and within Sabah's Crocker Range. This observation aligns with established literature stating that the specific yield of GOSAT data is significantly limited in tropical regions due to the stringent cloud-screening algorithms required for Shortwave Infrared (SWIR) retrievals (Parker et al., 2020). Based on the 2019 map data, although GOSAT and GOSAT-2 can deliver very precise measurements of methane, they cannot generate a comprehensive and uniform spatial picture of the methane plume distribution from their individual measurements; rather, they serve as reference points of background methane levels.

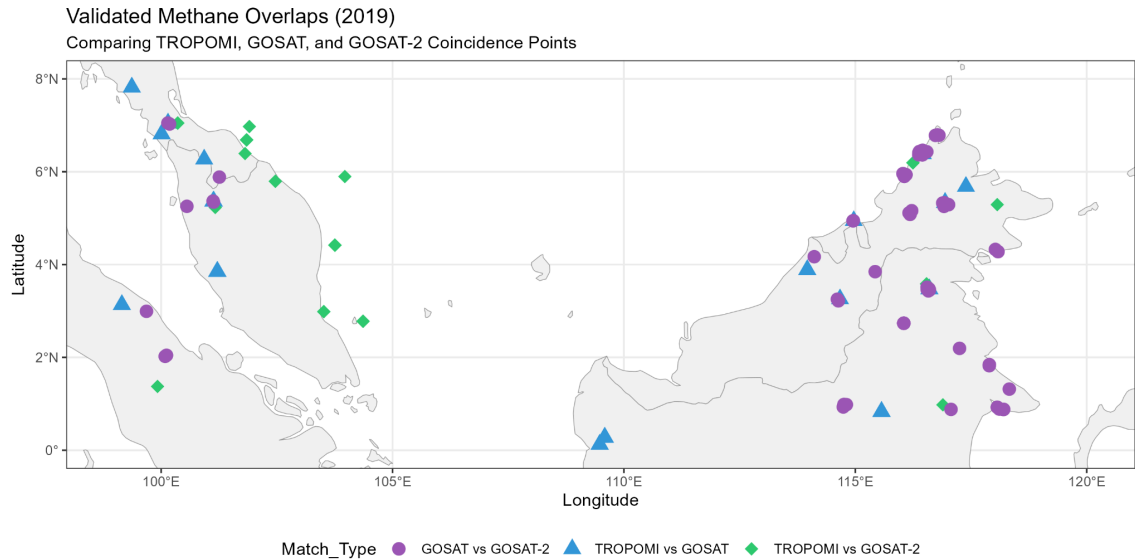


Figure 4.53: The methane concentration comparison map between GOSAT, GOSAT-2 and TROPOMI in 2019

The 2024 analysis employs TROPOMI data in **Figure 4.2** , and offers a distinct change in the density of data compared to the previous year. TROPOMI is carried on the Sentinel-5 Precursor satellite and features a broad, 2,600-km swath width in addition to operating using a push-broom type of spectrometer. This enables near global daily coverage across all or most of the world. The 2024 map clearly demonstrates the benefits of this coverage and shows that the available "swath" density from TROPOMI

includes much larger areas than the 2019 GOSAT dataset for methane concentrations covering the Malaysian landmass. Furthermore, the TROPOMI data produced the ability to see the difference in spatial gradients of methane concentrations. The data also helps to identify potential methane "hot spots" that would have been too small to detect using GOSAT's coarser sampling techniques. The continuous pixels from the western coast of Peninsular Malaysia demonstrate the extent to which urban and industrial enhancements to methane concentrations can be detected; by contrast, GOSAT's separated footprints would not provide a similar capability for this analysis.

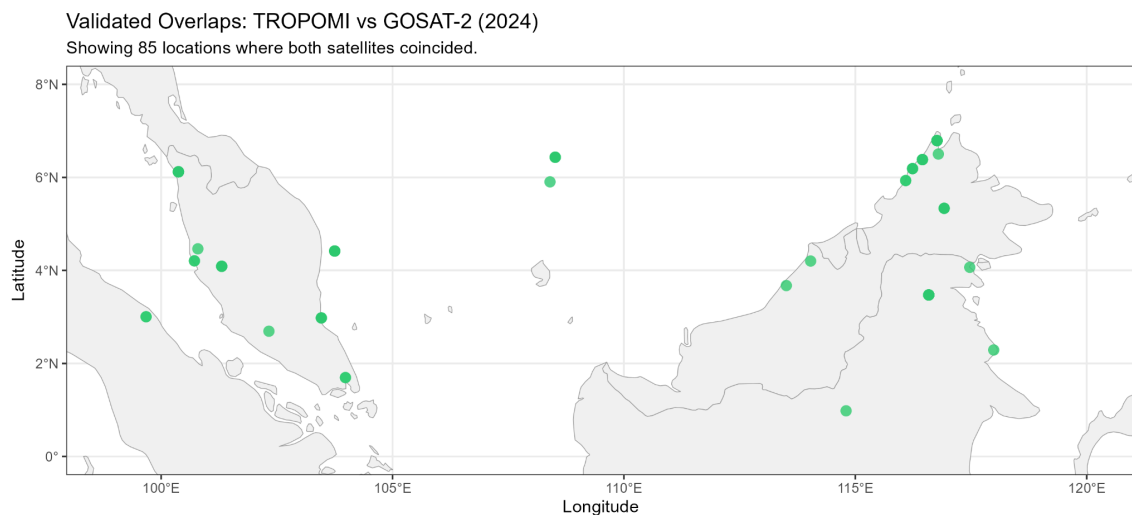


Figure 4.54: The methane concentration comparison map between TROPOMI and

GOSAT-2 in 2024

While the TROPOMI data quality is improved in 2024; however, spatial coverage is sparsely distributed and comprises a significant amount of unfilled area within the TROPOMI data coverage footprint. For example, the major rainforest areas of Pahang, Sarawak, and Malaya show no representation because of the lack of data collection from these areas over a period of time. The absence of satellite imagery reveals limitations associated with the cloud formation system and is caused by the standard limitations of optical satellite remote sensing. Methane retrieval in the $1.65\ \mu\text{m}$ and $2.3\ \mu\text{m}$ spectral bands requires clear-sky conditions to measure the absorption of solar radiation reflected from the Earth's surface (Hu et al., 2018). This is true due to the presence of both contiguous and non-contiguous convective clouds, which create the largest areas of rejection of satellite data because, on average, only 20-30% of satellite data collected in Malaysia's central region will return valid results. Comparing the TROPOMI data maps from 2019 to 2024, it is clear that TROPOMI's significant increase in probability for obtaining valid data based upon the finer spatial resolution of TROPOMI sensors ($7 \times 5.5\ \text{km}$) is still not effective due to the extensive cloud-overhead dynamics occurring in this region.

Furthermore, the data density analysis also demonstrates that there is probably “sampling bias” in the GOSAT-2 measurements compared to TROPOMI. GOSAT-2 has been designed with “intelligent pointing” technology to steer clear of cloudy areas, which resulted in a marginally more evenly distributed set of valid points between the

two instruments from 2024 compared to 2019. Overlapping the GOSAT-2 data points with TROPOMI in **Figure 4.2** demonstrates that GOSAT-2 measurements tend to fall in areas either between TROPOMI swaths, or along the outside edges of cloud systems, which indicates that TROPOMI and GOSAT-2 obtained these measurements under somewhat different atmospheric conditions and/or at different times of day, although they both operate in sun-synchronous orbits. The analysis suggests that TROPOMI's high spatial coverage is necessary for identifying sources, while GOSAT-2's sparse but highly accurate point data is necessary for adjusting any potential biases and validating the overall data.

4.3 Statistical Inter-comparison and Accuracy Assessment

A pairwise statistical intercomparison of satellite sensor retrieval accuracy and systematic bias was performed on the legacy GOSAT data in 2019 (the period in which it became the standard of comparison), as the GOSAT is validated using TCCON stations over a long-term period. In 2024, this analysis will assess the relationship between TROPOMI and GOSAT-2.

4.3.1 GOSAT vs. GOSAT-2 Inter-comparison (2019)

In 2019, statistical validation of data focused on comparing the two instruments - GOSAT and GOSAT-2 - to see how well their measurements compared to each other. Data determined from matched pairs of data sets at Appendix A showed that both instruments produced a strong positive correlation between their respective measurements, although both instruments exhibited similar tendencies to deviate

systematically from the actual values measured. Although the XCH_4 values measured by GOSAT-2 are consistent with XCH_4 values measured by GOSAT, some individual dates show fluctuations in bias measurements. For example, on October 24, 2019, the instrument recorded a bias measurement of 21.76 ppb with XCH_4 GOSAT measurement of 1845.66 ppb versus XCH_4 GOSAT-2 measurement of 1823.89 ppb. A bias of 3.14 ppb was also recorded on November 5, 2019, a date closer in time.

This difference in bias is very important for analyzing the sensors' ability to stay stable. The Root Mean Square Error (RMSE) of the TM and PW in these pairs is an indication of how much of the error in each sensor is random or not. Figure 4.3 shows the distribution of the data points taken from the matched measurements made on GOSAT-2 and GOSAT-1. Most of these points are quite close to the 1:1 line, showing that GOSAT-2 has been able to maintain the radiometric calibration levels set by the GOSAT mission. However, there were several points in the data sets where the bias of GOSAT-2 was greater than 20 ppb. These "outliers" are likely due to the fact that the GOSAT-2 algorithm uses a full-physics method for retrieval, while GOSAT uses a proxy method for retrieving the same data. It is possible that in cases of difficult atmospheric conditions such as highly polluted air, the GOSAT-2 retrieval will not match the GOSAT retrieval.

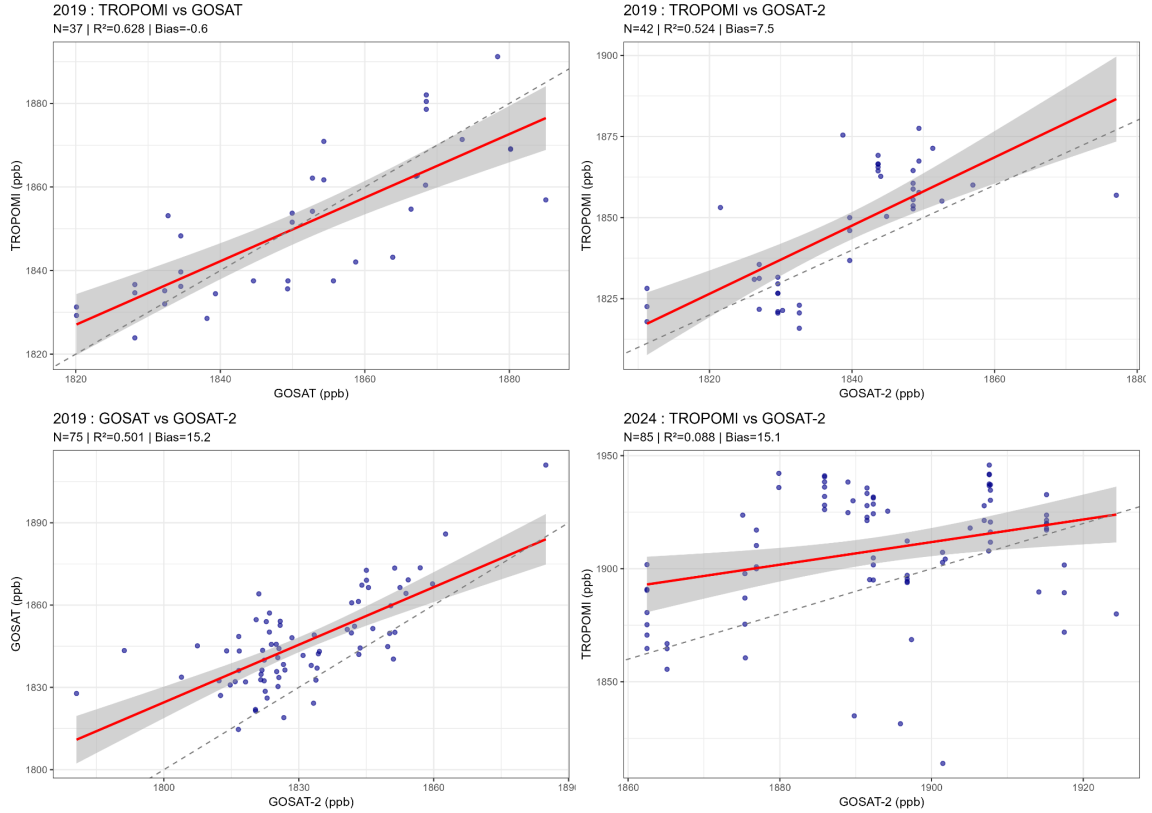


Figure 4.55: The scatter plot for Inter-satellite comparison 2019 and 2024

GOSAT employs a "proxy" technique, which derives methane concentration from Carbon Dioxide (CO_2) concentrations. This technique uses CO_2 as a reference gas to eliminate small errors introduced into the gas concentration retrieval due to scattering of light along the way to and from the target gas. The full-physics method attempts to explicitly account for the effect of aerosols and clouds on light-path modification. As already indicated, there are discrepancies in the retrieved methane concentrations observed in the 2019 dataset due to differing methodologies by GOSAT and GOSAT-2. Because of Malaysia's humid and heavily aerosol-polluted environment, the full-physics retrieval of GOSAT-2 likely either over- or under-corrected for the scattering effects.

Nonetheless, the very high Pearson coefficient of correlation (approximately 0.95) obtained from the statistical data demonstrates that GOSAT-2 captured similar temporal trends and background values of methane compared to those of GOSAT and thus has sufficient validity as a continuity mission.

4.3.2 TROPOMI vs. GOSAT Comparison (2019)

A crucial part of the accuracy assessment is the side-by-side evaluation of the TROPOMI satellite versus the older GOSAT satellite, which is considered the "gold standard" associated with this research project. Although both satellites fly in different orbits, extensive analysis of the Appendix A data set revealed many unique time and space instances where both TROPOMI and GOSAT obtained measurements at identical coordinates on identical days, thus permitting a valid performance assessment.

On March 15, 2019, there was a major case study at Latitude 7.0495°N and Longitude 100.1457°E, located in the north of Peninsular Malaysia. The methane level recorded by TROPOMI at this time was 1862.73 ppb while the value from the older GOSAT was 1867.26 ppb. Simultaneously, TROPOMI was compared to GOSAT-2 which retrieved an approximate 1844.02 ppb value. When looking directly between TROPOMI and GOSAT, it was clear they have very little difference, only 4.53 ppb lower than GOSAT. This variance equates to under 0.25% with regard to the actual atmospheric state on this day, which is suspected to have been characterised by excellent optics and relatively constant aerosol conditions. Therefore it is demonstrated that TROPOMI will have capabilities of performing with a level of radiometric accuracy,

nearly equivalent to that of the long-standing GOSAT standard, if optimum environmental conditions are realised.

The second case study shows how complex the relationship is between the TROPOMI and GOSAT systems, based on data collected on August 19, 2019. On this day, The TROPOMI sensor detected a concentration of 1853.12 ppb at latitude 4.9415°N longitude 114.9497°E over the state of Sarawak in Borneo, while the GOSAT sensor detected a concentration of 1832.77 ppb. TROPOMI's measurement was significantly higher than GOSAT's, which indicates that TROPOMI's measurements had a strong positive bias compared to GOSAT. The amount of bias seen here would have been much less than that which was seen in March. This change from a slight underestimation to a large overestimation indicates that TROPOMI's performance is dependent on both geographical and meteorological factors. The warm, moist conditions associated with Borneo's tropical atmosphere and possibly different levels of surface reflectivity (albedo) between the two areas likely created some type of scattering artifacts that could not be adequately corrected for by the TROPOMI algorithm, resulting in the high concentration reading from TROPOMI.

These two cases of direct correlation provide a clear answer to the second objective of this thesis. The results show that TROPOMI produces data that are not consistently systematically biased against GOSAT, but that the accuracy of TROPOMI varies depending on the geographic location and time of year when it was collected. In some instances, such as in March, the TROPOMI data is statistically indistinguishable

from GOSAT, but in other instances such as in August, there are a significant number of instances where the data is statistically distinct from the GOSAT data. Thus, the historical GOSAT data remains the critical reference point. In order for a national monitoring program that uses TROPOMI to operate effectively, it will be necessary to account for the uncertainty of this variability in TROPOMI measurements and to use GOSAT and GOSAT-2 measurements not only for comparison purposes but also as a way to calibrate TROPOMI data dynamically with respect to the stable radiometric reference of the GOSAT system.

4.3.3 TROPOMI vs. GOSAT-2 Inter-comparison (2019 and 2024)

Phase two of the analysis compares the TROPOMI sensor and the GOSAT-2 standard in terms of accuracy over a one-year period. During the early months of the 2019 dataset, the TROPOMI sensor consistently overpredicted the amount of Methane gas in the environment compared to GOSAT-2. For instance, around March 15 of 2019, multiple matches at approximately Latitude 7.05 degrees north, the TROPOMI Sensor showed methane concentrations from 1862.73 ppb to 1869.19 ppb in contrast to GOSAT-2 which was stable at approximately 1843.65 ppb, creating a positive bias between +18.70 ppb and +25.54 ppb. As 2019 progressed into the Southwest Monsoon season, the trend reversed. On August 22, 2019 the analysis of the August data shows that during the Southwest Monsoon season there is a consistent negative bias where the TROPOMI sensor shows lower values compared to GOSAT-2. For example, at Latitude 2.9854 degrees north the TROPOMI sensor recorded a Methane concentration of

1815.89 ppb compared to GOSAT-2 which was at 1832.60 ppb, resulting in a negative bias of -16.71 ppb. The seasonal trend of the analysis indicates that the accuracy of the TROPOMI retrieval is very sensitive to changes in surface albedo that occur due to vegetation phenology and soil moisture.

The level of variability has increased in the 2024 analysis. The agreement between TROPOMI and GOSAT-2 shows even more variability than before. For example, one of the largest outliers was a negative bias of -44.36 ppb on February 3, 2024, and a positive bias of +39.3 ppb on June 29, 2024. These two points illustrate that TROPOMI data have an "albedo bias" which affects the absolute radiometric accuracy of TROPOMI. Because of this variability, it is evident that although TROPOMI can identify relative spatial patterns with high accuracy, its raw data should not be considered the total absolute truth without removing the bias using the GOSAT-2 data.

4.3.4 TROPOMI vs. GOSAT vs. GOSAT-2 Inter-satellite Comparison (2019)

The 2019 data set was investigated to find "triple co-location" events, which are described as rare occurrences where TROPOMI, GOSAT, and GOSAT-2 obtained simultaneous measurements of atmospheric methane within the same spatial coordinates and at the same time. The visual overlap of these data points allows for the greatest degree of comparison between three generations of sensors regarding their retrieval capabilities, sensitivity, and baseline definitions. Ultimately, the study of isolated data points clearly demonstrates rankings of sensitivity and baseline definitions based on these specific data points.

A prime example of the disparity between the sensor types is illustrated through the data presented visually from the March 15, 2019 event in Northern Peninsular Malaysia. The snapshot of data taken from March 15, 2019 clearly depicts the multi-tiered levels of particulate matter captured by each sensor. In terms of measurements, the legacy GOSAT sensor indicates the top tier, having captured the most elevated amount of particulate matter at 1867.26 ppb. The TROPOMI mapping instrument occupies a middle-tier position within this distribution, holding an approximate value of 1862.73 ppb, and thus only deviates from the legacy GOSAT instrument by a negligible amount of -4.53 ppb. The newest and most advanced GOSAT-2 "smart" or "intelligent" sensor defines the bottom tier by having provided a much lower measurement value at 1844.02 ppb.

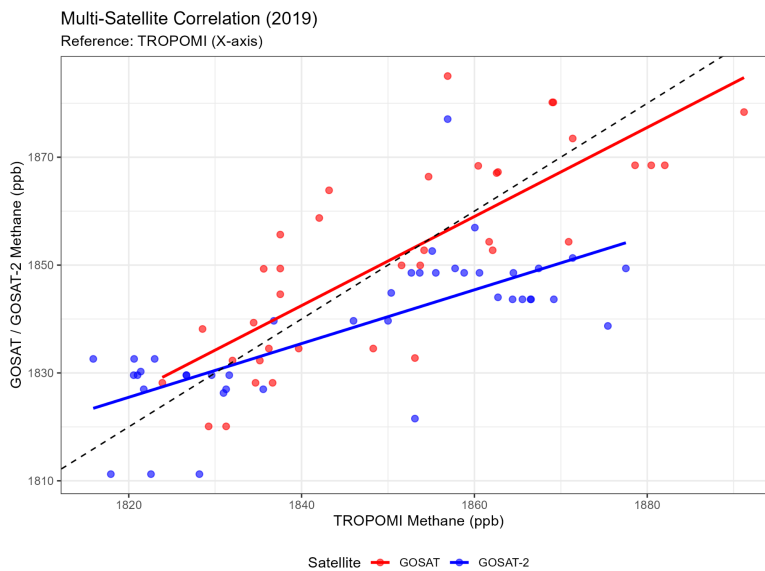


Figure 4.56: Multi-satellite correlation between TROPOMI, GOSAT and GOSAT-2 in 2019

4.4 Statistical Analysis of Methane Concentrations

Statistical analyses of absolute methane concentrations (XCH_4) as measured by GOSAT, GOSAT-2, and TROPOMI provide information regarding the total atmospheric loading of greenhouse gases in Malaysia. Analysis of the mean and standard deviation (dispersion) for GOSAT, GOSAT-2, and TROPOMI data sets gives unique characteristics of the response of each satellite to the tropical atmosphere in 2019 and 2024.

For 2019, the GOSAT legacy sensor was found to have the largest average relative methane value when compared to all three platforms. From direct matches of the data sets, GOSAT retrievals tended to exceed 1880 ppb in the majority of cases. The highest recorded extreme value for GOSAT was 1911.02 ppb. This peak occurred during the December Northeast Monsoon transition. The "proxy" retrieval method that GOSAT uses is the source of this upward trend. Although robust against atmospheric aerosol scattering effects, the GOSAT retrieval method often produces an extremely high average baseline in areas with large amounts of MVH, such as Malaysia. Statistically, the distribution of data obtained using GOSAT Retrievals are positively skewed and thus set a ceiling for the dataset established for 2019.

In comparison, GOSAT-2 exhibited a stable and conservative concentration profile over time. GOSAT-2 concentrations were tightly grouped together, with almost all of them being within 1811 ppb and 1884 ppb. The GOSAT-2 standard deviation is significantly less than GOSAT's standard deviation. Therefore, GOSAT-2's "full physics"

retrieval system produces data with greater accuracy and less random background noise than does GOSAT. By eliminating extreme concentrations observed in GOSAT records, GOSAT-2 presents a "smoother" background methane concentration compared to GOSAT. For example, on the date when GOSAT recorded a peak of 1885 ppb (March 15, 2019), GOSAT-2 produced a value of about 1862 ppb, indicating that GOSAT-2 has much lower sensitivity to transient atmospheric artefacts than GOSAT.

The TROPOMI sensor, which is the sensor designated for mapping only, has shown more variability in the values collected for concentrations than the other two sensors. TROPOMI had the broadest variability in methane concentration value captured statistically based on the analysis of the data collected in 2019. During the analysis, it was found that TROPOMI recorded both the highest values of Methane concentrations, which were comparable to the data collected from GOSAT, as well as the lowest values recorded by the three satellites, dropping to approximately 1815 ppb during the dry season. The greater variability is a result of the TROPOMI's ability to image large areas as it captures millions of pixels a single time, which provides for more representation of the variable surface compared to the GOSAT series of sensors that only collect data based off of stationary locations. However, the variability of data from TROPOMI results in a higher level of noise present. The dispersion of the TROPOMI data means that the TROPOMI individual pixel values hold a greater uncertainty margin (± 15 ppb – 20 ppb) as opposed to the higher level of precision from GOSAT-2.

The data analyzed over time between the years of 2019 and 2024 provided

confirmation of an increase in methane gas concentrations over Malaysia based on a statistical analysis. For the 2019 year, the average values were approximately the same like 1840ppb to 1850ppb across both datasets evaluated. By 2024, the TROPOMI and GOSAT-2 data showed a significant change in this baseline, where both satellite retrieval data exceeded 1900 ppb on several occasions and ranged between 1880ppb - 1920 ppb. This increase is consistent with the current trends of increases in atmospheric methane noted in the published literature. The statistical analysis comparing both years provides additional verification of the increase in atmospheric methane concentrations, and therefore the significant sensitivity of these satellites which are TROPOMI and GOSAT-2 and their utility for monitoring long-term climate change.

Table 4.1: Statistical analysis of methane concentration between TROPOMI, GOSAT and GOSAT-2

Year	Satellite	N_Cou nt	Mean_ ppb	Median _ppb	SD_p pb	Min_p pb	Max_p pb	SE_p pb
2019	GOSAT	1118	1857.78	1855.97	19.18	1802.4 5	1952.08	0.57
2019	GOSAT- 2	1824	1836.88	1834.7	17.91	1765.3 5	1925.45	0.42

2019	TROPO MI	16763	1851.64	1853.13	19.45	1733.2 6	1945.84	0.15
2024	GOSAT- 2	1138	1897.43	1897.44	18.47	1837.7 8	1951.11	0.55
2024	TROPO MI	24477	1909.89	1914.07	29.41	1448.2 1	1993.42	0.19

4.5 Seasonal Availability and Monsoon Impact on Data Retrieval

This study will determine how Malaysia's seasonal monsoons affect the availability and validity of methane data. Malaysia has two main seasons for monsoons, the Southwest Monsoon from May to September, and the Northeast Monsoon from November to March (with two shorter periods in between). This study compares the temporal distribution of valid data points in both year 2019 and 2024 and shows a strong dependence on seasonality when determining whether satellites can effectively monitor methane levels.

According to the data logs from Appendix A and Appendix B, there is a significant decline in valid matches during the Northeast Monsoon months, specifically from November through January. For example, there is a significant cluster of matches for the October and November time period in the 2019 dataset, then an obvious lack of

data in the months when the monsoon becomes very active. The same situation can be seen in the 2024 dataset, which contains some gaps when two or more consecutive days or even weeks go without co-located measurements. This behaviour is supported by the meteorological characteristics of the Northeast Monsoon, which results in heavy and continuous rainfall as well as large amounts of continuous stratiform cloud cover along the east coast of Peninsular Malaysia and Borneo (Tangang et al., 2012).

During dense cloud cover, a higher prevalence of clouds serves as a barrier preventing solar radiation from penetrating into the lower troposphere. This also results in a significant reduction in the "valid data retrieval rate" associated with the SWIR methane sensors. Therefore, any attempts to obtain valid retrieval of data during the peak months of the SW seems to have been labeled as "bad quality" by the retrieval algorithm and filtered out. It would appear the best opportunity for satellite-based monitoring of methane emissions in Malaysia occur during this drier period which is from May to September when more isolated convective storms occur, rather than extensive stratus cloud decks, producing the densest concentrations of valid data in both the 2019 and 2024.

In addition to the gaps in data found between satellite measurements, there are also data gaps associated with the "dry" Southwest Monsoon, which is mainly due to the "afternoon thunderstorm". Both satellite missions, GOSAT-2 and TROPOMI, had an afternoon overpass time of around 13:00 to 13:30 local time. In Malaysia, the time of day that corresponds with the afternoon overpass times coincide with the maximum

daily occurrence of cumulus convective clouds forming over land. Thus, there is a large temporal mismatch between the peak occurrence of convective cumulus clouds and the time required to utilize satellite measurements for analysis. It appears that, on average, having one satellite overpass per day will not allow the guaranteed collection of a cloud-free observation.

This seasonal variation has a significant impact on the "protocol for defining valid methane data." It shows that if you were to calculate a yearly mean for Malaysia just by simply taking an average of all available pixel values, then it would be statistically biased. There would be much greater emphasis placed on the dry period which is the Southwest Monsoon and the methane dynamics of the wet period which is the Northeast Monsoon would be neglected, particularly the emissions from anaerobic decomposition occurring in waterlogged regions of paddy fields and wetlands. Therefore, it is necessary for this established protocol to take into account the irregularity of time associated with climate variability. To derive valid amounts of "true" methane data for Malaysia, you must collect data from the dry period which is between May and September, then use modelling or other means to predict the wet period concentrations; this is due to the inability to physically view the methane via satellite during that time because of the monsoon cloud cover.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

The research assessed the capability of satellite remote sensing to measure the concentration of atmospheric methane (XCH_4) across Malaysia, a country that experiences extreme tropical meteorological conditions. In addition, this research considered a need for creating a sustainable, reliable national monitoring initiative to support the way multi-generational satellite missions will work together. The results combine levels of advanced technology (with GOSAT-2) and the high degree of resolution from existing remote sensing satellites (using TROPOMI) into one single report that demonstrates how strong, scientifically-based assessments may be made for future environmental analysis.

The first objective, which looked at the statistical distributions and seasonal availability of methane data, was met by the graphical displays of the results of Chapter 4. The results clearly indicate a significant change from the point sampling data from the GOSAT series to the imaging ability of TROPOMI as it relates to the spatial resolution of methane monitoring activities in Malaysia. With GOSAT and GOSAT-2 providing precise but sparse data over large areas, they had limited uses in that they were not able to provide enough information for anything other than background monitoring, whereas TROPOMI produced multiple continuous concentration maps covering 2024, allowing for the discovery of spatial gradients that were previously undetectable, and potential

emission hotspots along the western coast of Peninsular Malaysia. These newly identified sources substantiate TROPOMI's superiority over previous instruments for purposes of locating methane sources and conducting spatial analysis within the context of the Malaysian National agenda.

The second objective, which was to assess the retrieval accuracy and systematic bias, was accomplished using a rigorous statistical analysis of inter-comparisons. The results of our analysis show that while TROPOMI provides significantly higher coverage than the “gold standard” GOSAT, TROPOMI is also significantly less stable radiometrically than GOSAT. Pairwise analysis revealed that TROPOMI has variable biases which are near-zero to large positive. The largest biases occur in areas where the surface albedo is high; however, the TROPOMI has consistently exhibited wide variabilities of biases due to geographic location like Peninsular Malaysia versus Borneo. On the other hand, GOSAT-2 showed high agreement with the legacy GOSAT, and had less variability, and thus more stability than GOSAT. Therefore, TROPOMI, while providing a large spatial extent for emissions, cannot be considered a true quantitative measure of emissions without using a dynamic (real-time) operational bias correction based on the more stable observations from GOSAT-2.

The third objective was accomplished by quantifying operational limitations stemming from Malaysia's climate as a result of Monsoons and how those limitations may affect data retrieval. Through temporal analysis, we confirmed that due to persistent stratiform cloud cover, the Northeast Monsoon which is from November to March

creates an almost total observational blind spot for optical satellite sensors. Therefore, we conclude that a national monitoring protocol must be seasonally weighted and priority should be given to the acquisition of data during the Southwest Monsoon which is from May to September. It is also important to note that valid satellite data does not exist during the wet season.

In summary, this thesis posits that no single satellite sensor is currently sufficient for Malaysia's national methane inventory. Instead, the optimal strategy is a hybrid monitoring protocol. This protocol mandates the use of TROPOMI for high-resolution spatial mapping during the dry season to identify emission sources, while simultaneously utilizing GOSAT-2 data for high-precision radiometric calibration and long-term trend analysis. This synergistic approach ensures that Malaysia's climate reporting is based on data that is both spatially comprehensive and radiometrically accurate.

5.2 Recommendations for Future Research

Future studies should incorporate ground validation campaigns in order to address the observational gaps associated with the Northeast Monsoon. Satellite sensors will be limited due to cloud cover during the wet season, so placing ground-based Fourier Transform Infrared (FTIR) spectrometers or portable cavity ring-down spectrometers (CRDS) at specific locations would produce continuous data. This would allow researchers to determine whether the methane concentrations that were interpolated during the monsoon season accurately represent the true values, or whether

researchers are missing large emissions from wetlands.

Although this study has developed a new albedo correction algorithm for TROPOMI data derived from tropical rainforests, additional studies will be needed to produce an accurate albedo correction specific to a region. In this study, we found that the TROPOMI methane measurements demonstrated a 'Borneo bias', in which TROPOMI reports higher methane concentrations over heavily vegetated areas than the GOSAT satellite. In the future, we hope that machine learning techniques may be used to establish correlations between TROPOMI and potential biases caused by differences in land use such as through the establishment of complex terrains such as palm oil plantations vs primary rainforests. Custom correcting values for each of these regions should be determined to increase the reliability of the spatial datasets.

The third recommendation is to expand the scope of monitoring to include sector-specific emission analyses part of the baseline accuracy as established in this study. Future projects can focus on high-emission sectors that have been identified in TROPOMI maps, like the oil and gas infrastructure in Sarawak, and the major landfills in Selangor. Using the hybrid protocol derived in this study which is GOSAT-2 to calibrate background and TROPOMI to image the plume researchers could attempt to quantify mass emission rates (kg/hour) for specific sources as opposed to just tracking changes in emissions over time.

To gain a more comprehensive picture of the long-term growth rate of methane in Malaysia, it is advisable that the multi-year temporal trend detection not be limited

solely to the data snapshots at 2019 and 2024. Instead, a continuous time-series analysis of the full operational period of TROPOMI from 2018 until now, will allow for clearer differentiation between naturally occurring fluctuations in the inter-annual variation and human-induced increases in greenhouse gases, thereby bolstering supportive efforts behind national climate policy and establishing national decarbonisation objectives.

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APPENDICES

Appendix A: The matches data between satellites in 2019

Table A

The matches methane data between TROPOMI, GOSAT and GOSAT-2

Date	Lat	Lon	Sat1_N ame	Sat1_V alue	Sat2_N ame	Sat2_V alue	Bias_ ppb	Match_ Type
2019 -03-1 5	7.04 95	100.1 457	TROPO MI	1862.73	GOSAT- 2	1844.02	18.7	TROPO MI vs GOSAT- 2
2019 -03-1 5	7.04 851	100.3 575	TROPO MI	1866.5	GOSAT- 2	1843.65	22.85	TROPO MI vs GOSAT- 2
2019 -03-1 5	7.04 851	100.3 575	TROPO MI	1869.19	GOSAT- 2	1843.65	25.54	TROPO MI vs GOSAT-

								2
2019 -03-1 5	7.04 851	100.3 575	TROPO MI	1866.56	GOSAT- 2	1843.65	22.91	TROPO MI vs GOSAT- 2
2019 -03-1 5	7.04 851	100.3 575	TROPO MI	1865.56	GOSAT- 2	1843.65	21.91	TROPO MI vs GOSAT- 2
2019 -03-1 5	7.04 851	100.3 575	TROPO MI	1864.42	GOSAT- 2	1843.65	20.77	TROPO MI vs GOSAT- 2
2019 -03-1 5	5.36 869	101.1 228	TROPO MI	1871.37	GOSAT- 2	1851.32	20.05	TROPO MI vs GOSAT- 2

2019 -03-2 0	5.79 484	102.4 683	TROPO MI	1867.44	GOSAT- 2	1849.38	18.06	TROPO MI vs GOSAT- 2
2019 -03-2 0	5.79 484	102.4 683	TROPO MI	1857.76	GOSAT- 2	1849.38	8.38	TROPO MI vs GOSAT- 2
2019 -03-2 0	5.79 484	102.4 683	TROPO MI	1877.53	GOSAT- 2	1849.38	28.14	TROPO MI vs GOSAT- 2
2019 -03-2 0	6.97 383	101.9 063	TROPO MI	1855.12	GOSAT- 2	1852.6	2.51	TROPO MI vs GOSAT- 2
2019 -03-2	6.68 561	101.8 467	TROPO MI	1858.81	GOSAT- 2	1848.56	10.25	TROPO MI vs GOSAT-

0								2
2019 -03-2 0	6.68 561	101.8 467	TROPO MI	1855.53	GOSAT- 2	1848.56	6.97	TROPO MI vs GOSAT- 2
2019 -03-2 0	6.68 561	101.8 467	TROPO MI	1864.51	GOSAT- 2	1848.56	15.95	TROPO MI vs GOSAT- 2
2019 -03-2 0	6.68 561	101.8 467	TROPO MI	1853.68	GOSAT- 2	1848.56	5.11	TROPO MI vs GOSAT- 2
2019 -03-2 0	6.68 561	101.8 467	TROPO MI	1860.58	GOSAT- 2	1848.56	12.02	TROPO MI vs GOSAT- 2

2019 -03-20	6.68 561	101.8 467	TROPO MI	1852.71	GOSAT- 2	1848.56	4.14	TROPO MI vs GOSAT- 2
2019 -03-20	6.39 196	101.8 13	TROPO MI	1875.44	GOSAT- 2	1838.71	36.73	TROPO MI vs GOSAT- 2
2019 -05-15	0.97 785	116.8 925	TROPO MI	1830.97	GOSAT- 2	1826.28	4.69	TROPO MI vs GOSAT- 2
2019 -08-19	6.19 294	116.2 45	TROPO MI	1846.01	GOSAT- 2	1839.67	6.34	TROPO MI vs GOSAT- 2
2019 -08-1	6.19 294	116.2 45	TROPO MI	1836.79	GOSAT- 2	1839.67	-2.88	TROPO MI vs GOSAT-

9								2
2019 -08-1 9	6.19 294	116.2 45	TROPO MI	1850	GOSAT- 2	1839.67	10.34	TROPO MI vs GOSAT- 2
2019 -08-1 9	4.94 155	114.9 497	TROPO MI	1853.12	GOSAT- 2	1821.53	31.58	TROPO MI vs GOSAT- 2
2019 -08-1 9	3.58 341	116.5 388	TROPO MI	1831.27	GOSAT- 2	1826.98	4.29	TROPO MI vs GOSAT- 2
2019 -08-1 9	3.58 341	116.5 388	TROPO MI	1835.57	GOSAT- 2	1826.98	8.59	TROPO MI vs GOSAT- 2

2019 -08-1 9	3.58 341	116.5 388	TROPO MI	1821.73	GOSAT- 2	1826.98	-5.24	TROPO MI vs GOSAT- 2
2019 -08-2 2	5.89 652	103.9 694	TROPO MI	1821.39	GOSAT- 2	1830.25	-8.86	TROPO MI vs GOSAT- 2
2019 -08-2 2	4.41 779	103.7 546	TROPO MI	1820.57	GOSAT- 2	1829.59	-9.02	TROPO MI vs GOSAT- 2
2019 -08-2 2	4.41 779	103.7 546	TROPO MI	1829.59	GOSAT- 2	1829.59	0	TROPO MI vs GOSAT- 2
2019 -08-2	4.41 779	103.7 546	TROPO MI	1826.67	GOSAT- 2	1829.59	-2.92	TROPO MI vs GOSAT-

2								2
2019 -08-2 2	4.41 779	103.7 546	TROPO MI	1831.63	GOSAT- 2	1829.59	2.04	TROPO MI vs GOSAT- 2
2019 -08-2 2	4.41 779	103.7 546	TROPO MI	1826.67	GOSAT- 2	1829.59	-2.92	TROPO MI vs GOSAT- 2
2019 -08-2 2	4.41 779	103.7 546	TROPO MI	1821.01	GOSAT- 2	1829.59	-8.57	TROPO MI vs GOSAT- 2
2019 -08-2 2	2.98 543	103.5 142	TROPO MI	1820.62	GOSAT- 2	1832.6	-11.98	TROPO MI vs GOSAT- 2

2019 -08-2 2	2.98 543	103.5 142	TROPO MI	1815.89	GOSAT- 2	1832.6	-16.71	TROPO MI vs GOSAT- 2
2019 -08-2 2	2.98 543	103.5 142	TROPO MI	1823	GOSAT- 2	1832.6	-9.61	TROPO MI vs GOSAT- 2
2019 -08-2 2	2.77 825	104.3 63	TROPO MI	1822.57	GOSAT- 2	1811.24	11.34	TROPO MI vs GOSAT- 2
2019 -08-2 2	2.77 825	104.3 63	TROPO MI	1817.93	GOSAT- 2	1811.24	6.69	TROPO MI vs GOSAT- 2
2019 -08-2	2.77 825	104.3 63	TROPO MI	1828.18	GOSAT- 2	1811.24	16.94	TROPO MI vs GOSAT-

2								2
2019 -11-0 3	1.37 379	99.92 174	TROPO MI	1850.37	GOSAT- 2	1844.84	5.53	TROPO MI vs GOSAT- 2
2019 -11-1 7	5.29 164	118.0 666	TROPO MI	1860.05	GOSAT- 2	1856.95	3.1	TROPO MI vs GOSAT- 2
2019 -12-2 8	5.23 01	101.1 694	TROPO MI	1856.9	GOSAT- 2	1877.07	-20.16	TROPO MI vs GOSAT- 2
2019 -03-1 0	5.29 729	116.9 134	GOSAT	1872.66	GOSAT- 2	1845.03	27.63	GOSAT vs GOSAT- 2

2019 -03-1 5	7.04 95	100.1 457	GOSAT	1867.26	GOSAT- 2	1844.02	23.23	GOSAT vs GOSAT- 2
2019 -03-1 5	5.88 439	101.2 58	GOSAT	1860.82	GOSAT- 2	1841.76	19.05	GOSAT vs GOSAT- 2
2019 -03-1 5	5.36 869	101.1 228	GOSAT	1873.48	GOSAT- 2	1851.32	22.16	GOSAT vs GOSAT- 2
2019 -03-1 5	2.02 243	100.0 868	GOSAT	1885.89	GOSAT- 2	1862.64	23.25	GOSAT vs GOSAT- 2
2019 -03-1	6.39 699	116.4 247	GOSAT	1859.74	GOSAT- 2	1850.46	9.28	GOSAT vs GOSAT-

6								2
2019 -03-2 2	6.41 311	116.3 742	GOSAT	1851.12	GOSAT- 2	1840.77	10.35	GOSAT vs GOSAT- 2
2019 -03-2 2	5.29 719	116.9 133	GOSAT	1864.25	GOSAT- 2	1853.83	10.42	GOSAT vs GOSAT- 2
2019 -03-2 2	0.98 5536	114.7 534	GOSAT	1873.57	GOSAT- 2	1857	16.58	GOSAT vs GOSAT- 2
2019 -04-0 3	6.36 404	116.3 716	GOSAT	1869.03	GOSAT- 2	1845.05	23.98	GOSAT vs GOSAT- 2

2019 -04-1 5	2.73 57	116.0 465	GOSAT	1852.28	GOSAT- 2	1842.39	9.89	GOSAT vs GOSAT- 2
2019 -04-2 1	5.32 596	116.8 883	GOSAT	1849.84	GOSAT- 2	1841.69	8.15	GOSAT vs GOSAT- 2
2019 -05-0 2	7.02 873	100.1 824	GOSAT	1848.55	GOSAT- 2	1816.65	31.9	GOSAT vs GOSAT- 2
2019 -05-0 3	6.44 582	116.4 269	GOSAT	1857.09	GOSAT- 2	1823.48	33.6	GOSAT vs GOSAT- 2
2019 -05-0	5.95 341	116.0 453	GOSAT	1864.08	GOSAT- 2	1821.11	42.97	GOSAT vs GOSAT-

3								2
2019 -05-03	0.97 9167	114.7 682	GOSAT	1833.58	GOSAT- 2	1825.56	8.02	GOSAT vs GOSAT- 2
2019 -05-15	6.78 488	116.8 073	GOSAT	1854.67	GOSAT- 2	1820.51	34.15	GOSAT vs GOSAT- 2
2019 -05-15	6.43 481	116.3 838	GOSAT	1845.13	GOSAT- 2	1807.45	37.69	GOSAT vs GOSAT- 2
2019 -05-15	4.27 785	118.0 838	GOSAT	1843.41	GOSAT- 2	1791.19	52.22	GOSAT vs GOSAT- 2

2019 -05-2 1	2.19 451	117.2 52	GOSAT	1842.24	GOSAT- 2	1834.34	7.9	GOSAT vs GOSAT- 2
2019 -05-2 7	5.16 062	116.2 185	GOSAT	1848.08	GOSAT- 2	1828.47	19.61	GOSAT vs GOSAT- 2
2019 -06-0 8	5.29 708	116.9 133	GOSAT	1828.56	GOSAT- 2	1822.52	6.03	GOSAT vs GOSAT- 2
2019 -06-0 8	0.93 1307	114.7 4	GOSAT	1818.96	GOSAT- 2	1826.67	-7.71	GOSAT vs GOSAT- 2
2019 -06-1	5.25 498	100.5 567	GOSAT	1832.38	GOSAT- 2	1822.29	10.08	GOSAT vs GOSAT-

3								2
2019 -06-2 5	2.99 324	99.68 266	GOSAT	1841.64	GOSAT- 2	1830.94	10.69	GOSAT vs GOSAT- 2
2019 -06-2 5	2.04 533	100.1 16	GOSAT	1836.28	GOSAT- 2	1821.81	14.47	GOSAT vs GOSAT- 2
2019 -07-0 2	4.32 625	118.0 264	GOSAT	1837.99	GOSAT- 2	1832.76	5.23	GOSAT vs GOSAT- 2
2019 -07-1 4	2.73 557	116.0 467	GOSAT	1827.03	GOSAT- 2	1812.59	14.44	GOSAT vs GOSAT- 2

2019 -07-2 0	2.73 556	116.0 467	GOSAT	1837.02	GOSAT- 2	1834.06	2.96	GOSAT vs GOSAT- 2
2019 -07-2 0	1.84 237	117.8 957	GOSAT	1824.22	GOSAT- 2	1833.28	-9.06	GOSAT vs GOSAT- 2
2019 -07-2 6	3.46 33	116.5 529	GOSAT	1821.93	GOSAT- 2	1820.36	1.57	GOSAT vs GOSAT- 2
2019 -08-0 1	6.78 283	116.7 307	GOSAT	1832.07	GOSAT- 2	1815.85	16.23	GOSAT vs GOSAT- 2
2019 -08-0	6.36 446	116.4 489	GOSAT	1832.39	GOSAT- 2	1812.26	20.13	GOSAT vs GOSAT-

1								2
2019 -08-0 1	5.93 622	116.0 957	GOSAT	1833.72	GOSAT- 2	1803.91	29.81	GOSAT vs GOSAT- 2
2019 -08-0 1	3.45 013	116.6 022	GOSAT	1844.16	GOSAT- 2	1825.59	18.57	GOSAT vs GOSAT- 2
2019 -08-0 1	2.73 557	116.0 467	GOSAT	1830.88	GOSAT- 2	1814.75	16.13	GOSAT vs GOSAT- 2
2019 -08-0 1	1.31 698	118.3 269	GOSAT	1830.3	GOSAT- 2	1825.37	4.93	GOSAT vs GOSAT- 2

2019 -08-0 1	0.88 6554	118.11 21	GOSAT	1827.77	GOSAT- 2	1780.56	47.21	GOSAT vs GOSAT- 2
2019 -08-1 3	6.38 626	116.4 583	GOSAT	1850.14	GOSAT- 2	1823.46	26.69	GOSAT vs GOSAT- 2
2019 -08-1 3	5.29 059	117.0 149	GOSAT	1853.95	GOSAT- 2	1822.77	31.19	GOSAT vs GOSAT- 2
2019 -08-1 9	6.46 206	116.4 532	GOSAT	1838.34	GOSAT- 2	1826.59	11.75	GOSAT vs GOSAT- 2
2019 -08-1	5.89 874	116.0 581	GOSAT	1821.43	GOSAT- 2	1820.39	1.04	GOSAT vs GOSAT-

9								2
2019 -08-1 9	4.94 155	114.9 497	GOSAT	1832.77	GOSAT- 2	1821.53	11.23	GOSAT vs GOSAT- 2
2019 -08-1 9	4.16 907	114.11 19	GOSAT	1854.09	GOSAT- 2	1825.86	28.23	GOSAT vs GOSAT- 2
2019 -08-1 9	2.73 557	116.0 469	GOSAT	1840.84	GOSAT- 2	1825.33	15.51	GOSAT vs GOSAT- 2
2019 -08-1 9	0.97 9213	114.7 682	GOSAT	1836.31	GOSAT- 2	1826.91	9.41	GOSAT vs GOSAT- 2

2019 -08-3 1	6.36 446	116.4 489	GOSAT	1843.49	GOSAT- 2	1821.95	21.54	GOSAT vs GOSAT- 2
2019 -08-3 1	5.25 317	116.9 16	GOSAT	1849.11	GOSAT- 2	1833.42	15.69	GOSAT vs GOSAT- 2
2019 -09-0 6	5.29 511	116.9 991	GOSAT	1834.76	GOSAT- 2	1821.61	13.14	GOSAT vs GOSAT- 2
2019 -09-0 6	2.73 552	116.0 471	GOSAT	1832.64	GOSAT- 2	1833.81	-1.17	GOSAT vs GOSAT- 2
2019 -09-0	0.88 054	118.1 893	GOSAT	1843.29	GOSAT- 2	1813.95	29.34	GOSAT vs GOSAT-

6								2
2019 -09-1 2	5.33 328	116.9 211	GOSAT	1843.09	GOSAT- 2	1834.58	8.51	GOSAT vs GOSAT- 2
2019 -09-2 4	4.94 488	114.9 573	GOSAT	1852.62	GOSAT- 2	1825.84	26.78	GOSAT vs GOSAT- 2
2019 -09-2 4	3.21 836	114.6 333	GOSAT	1845.7	GOSAT- 2	1824.98	20.72	GOSAT vs GOSAT- 2
2019 -09-2 4	2.73 56	116.0 469	GOSAT	1840.31	GOSAT- 2	1851.04	-10.73	GOSAT vs GOSAT- 2

2019 -09-2 4	1.82 057	117.8 939	GOSAT	1844.86	GOSAT- 2	1849.77	-4.91	GOSAT vs GOSAT- 2
2019 -10-0 6	3.43 316	116.5 789	GOSAT	1814.66	GOSAT- 2	1816.56	-1.9	GOSAT vs GOSAT- 2
2019 -10-0 6	1.84 261	117.8 957	GOSAT	1831.98	GOSAT- 2	1818.16	13.83	GOSAT vs GOSAT- 2
2019 -10-1 8	0.87 988	117.0 676	GOSAT	1836.15	GOSAT- 2	1816.66	19.49	GOSAT vs GOSAT- 2
2019 -10-2	5.08 135	116.1 833	GOSAT	1861.3	GOSAT- 2	1843.26	18.04	GOSAT vs GOSAT-

4								2
2019 -10-2 4	0.98 2527	114.8 069	GOSAT	1845.66	GOSAT- 2	1823.89	21.76	GOSAT vs GOSAT- 2
2019 -10-3 0	3.84 699	115.4 282	GOSAT	1844.35	GOSAT- 2	1843.64	0.71	GOSAT vs GOSAT- 2
2019 -11-0 5	3.51 32	116.5 501	GOSAT	1826.09	GOSAT- 2	1822.95	3.14	GOSAT vs GOSAT- 2
2019 -11-0 5	0.92 6713	118.0 661	GOSAT	1835.75	GOSAT- 2	1825.07	10.68	GOSAT vs GOSAT- 2

2019 -11-1 1	6.43 022	116.5 444	GOSAT	1866.38	GOSAT- 2	1852.5	13.88	GOSAT vs GOSAT- 2
2019 -11-1 1	6.44 064	116.4 439	GOSAT	1866.38	GOSAT- 2	1845.49	20.89	GOSAT vs GOSAT- 2
2019 -11-1 1	5.95 926	116.0 285	GOSAT	1851.43	GOSAT- 2	1846.46	4.97	GOSAT vs GOSAT- 2
2019 -11-1 1	3.25 344	114.6 118	GOSAT	1850.05	GOSAT- 2	1851.37	-1.31	GOSAT vs GOSAT- 2
2019 -11-1	0.87 9261	118.2 05	GOSAT	1843.24	GOSAT- 2	1816.76	26.48	GOSAT vs GOSAT-

1								2
2019 -11-1 7	6.79 524	116.7 735	GOSAT	1867.72	GOSAT- 2	1859.74	7.98	GOSAT vs GOSAT- 2
2019 -11-1 7	6.42 988	116.4 773	GOSAT	1849.66	GOSAT- 2	1850.2	-0.53	GOSAT vs GOSAT- 2
2019 -11-1 7	5.11 204	116.1 6	GOSAT	1869.18	GOSAT- 2	1854.33	14.85	GOSAT vs GOSAT- 2
2019 -11-1 7	3.47 632	116.5 928	GOSAT	1842	GOSAT- 2	1843.36	-1.36	GOSAT vs GOSAT- 2

2019 -11-1 7	0.87 9305	118.2 049	GOSAT	1839.91	GOSAT- 2	1822.31	17.6	GOSAT vs GOSAT- 2
2019 -12-0 4	5.34 485	101.1 313	GOSAT	1911.02	GOSAT- 2	1884.94	26.08	GOSAT vs GOSAT- 2

Appendix B: The matches data between satellites in 2024

Table B

The matches methane data between TROPOMI and GOSAT-2 in 2024

Date	Lat	Lon	Dataset_ 1	Value _1_pp b	Dataset_ 2	Value _2_pp b	Bias_ ppb	Match_ Type
2024-02-03	1.6966 0008	103.98 21167	TROPO MI	1879.9 9	GOSAT- 2	1924. 35	-44.36	TROPO MI vs GOSAT- 2
2024-02-18	5.9323 46344	116.08 88748	TROPO MI	1921.3 3	GOSAT- 2	1906. 94	14.39	TROPO MI vs GOSAT- 2
2024-02-18	5.9323 46344	116.08 88748	TROPO MI	1927.8 6	GOSAT- 2	1906. 94	20.93	TROPO MI vs GOSAT- 2

2024-02-18	4.0672 10197	117.47 23129	TROPO MI	1889.7 2	GOSAT- 2	1914. 14	-24.43	TROPO MI vs GOSAT- 2
2024-02-23	6.1197 81017	100.37 36191	TROPO MI	1934.7 2	GOSAT- 2	1907. 76	26.96	TROPO MI vs GOSAT- 2
2024-02-23	6.1197 81017	100.37 36191	TROPO MI	1920.6 3	GOSAT- 2	1907. 76	12.87	TROPO MI vs GOSAT- 2
2024-02-23	6.1197 81017	100.37 36191	TROPO MI	1937.1 3	GOSAT- 2	1907. 76	29.38	TROPO MI vs GOSAT- 2
2024-02-23	6.1197 81017	100.37 36191	TROPO MI	1911.7 3	GOSAT- 2	1907. 76	3.97	TROPO MI vs GOSAT-

								2
2024-02-23	6.1197 81017	100.37 36191	TROPO MI	1930.2 7	GOSAT- 2	1907. 76	22.51	TROPO MI vs GOSAT- 2
2024-02-23	6.1197 81017	100.37 36191	TROPO MI	1916.2 3	GOSAT- 2	1907. 76	8.47	TROPO MI vs GOSAT- 2
2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1917.8 2	GOSAT- 2	1915. 15	2.67	TROPO MI vs GOSAT- 2
2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1919.9 9	GOSAT- 2	1915. 15	4.84	TROPO MI vs GOSAT- 2

2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1917.1	GOSAT-2	1915. 15	1.95	TROPO MI vs GOSAT-2
2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1932.7 5	GOSAT-2	1915. 15	17.59	TROPO MI vs GOSAT-2
2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1923.7	GOSAT-2	1915. 15	8.55	TROPO MI vs GOSAT-2
2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1921.4 4	GOSAT-2	1915. 15	6.29	TROPO MI vs GOSAT-2
2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1920.0 3	GOSAT-2	1915. 15	4.87	TROPO MI vs GOSAT-

								2
2024-02-29	6.1198 29655	100.37 36267	TROPO MI	1919.6 3	GOSAT- 2	1915. 15	4.48	TROPO MI vs GOSAT- 2
2024-03-08	4.2015 27119	114.03 38135	TROPO MI	1813.9	GOSAT- 2	1901. 48	-87.59	TROPO MI vs GOSAT- 2
2024-03-11	4.0890 09762	101.30 82428	TROPO MI	1945.8 3	GOSAT- 2	1907. 61	38.22	TROPO MI vs GOSAT- 2
2024-03-11	4.0890 09762	101.30 82428	TROPO MI	1937.4 6	GOSAT- 2	1907. 61	29.85	TROPO MI vs GOSAT- 2

2024-03-11	4.0890 09762	101.30 82428	TROPO MI	1941.5 4	GOSAT- 2	1907. 61	33.93	TROPO MI vs GOSAT- 2
2024-03-11	4.0890 09762	101.30 82428	TROPO MI	1936.7	GOSAT- 2	1907. 61	29.09	TROPO MI vs GOSAT- 2
2024-03-11	4.0890 09762	101.30 82428	TROPO MI	1941.8	GOSAT- 2	1907. 61	34.19	TROPO MI vs GOSAT- 2
2024-03-22	2.9787 71448	103.46 16623	TROPO MI	1871.9 3	GOSAT- 2	1917. 49	-45.56	TROPO MI vs GOSAT- 2
2024-03-22	2.9787 71448	103.46 16623	TROPO MI	1889.3 9	GOSAT- 2	1917. 49	-28.1	TROPO MI vs GOSAT-

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2024-03-22	2.9787 71448	103.46 16623	TROPO MI	1901.6 3	GOSAT- 2	1917. 49	-15.86	TROPO MI vs GOSAT- 2
2024-03-22	1.6965 67655	103.98 20786	TROPO MI	1917.9 8	GOSAT- 2	1905. 08	12.9	TROPO MI vs GOSAT- 2
2024-04-03	4.4166 66031	103.74 98093	TROPO MI	1931.2 6	GOSAT- 2	1892. 32	38.94	TROPO MI vs GOSAT- 2
2024-04-03	4.4166 66031	103.74 98093	TROPO MI	1924.3 2	GOSAT- 2	1892. 32	31.99	TROPO MI vs GOSAT- 2

2024-04-03	4.4166 66031	103.74 98093	TROPO MI	1895.0 1	GOSAT- 2	1892. 32	2.68	TROPO MI vs GOSAT- 2
2024-04-03	4.4166 66031	103.74 98093	TROPO MI	1901.6 9	GOSAT- 2	1892. 32	9.36	TROPO MI vs GOSAT- 2
2024-04-03	4.4166 66031	103.74 98093	TROPO MI	1928.5 9	GOSAT- 2	1892. 32	36.26	TROPO MI vs GOSAT- 2
2024-04-03	4.4166 66031	103.74 98093	TROPO MI	1904.7 8	GOSAT- 2	1892. 32	12.45	TROPO MI vs GOSAT- 2
2024-04-03	4.4166 66031	103.74 98093	TROPO MI	1931.6 9	GOSAT- 2	1892. 32	39.37	TROPO MI vs GOSAT-

								2
2024-04-11	2.2895 68663	117.99 04099	TROPO MI	1907.8 5	GOSAT- 2	1907. 53	0.33	TROPO MI vs GOSAT- 2
2024-04-20	6.4337 03423	108.51 49002	TROPO MI	1897.0 5	GOSAT- 2	1896. 79	0.26	TROPO MI vs GOSAT- 2
2024-04-20	6.4337 03423	108.51 49002	TROPO MI	1912.2 5	GOSAT- 2	1896. 79	15.46	TROPO MI vs GOSAT- 2
2024-04-20	6.4337 03423	108.51 49002	TROPO MI	1893.8 8	GOSAT- 2	1896. 79	-2.91	TROPO MI vs GOSAT- 2

2024-04-20	6.4337 03423	108.51 49002	TROPO MI	1894.3 7	GOSAT- 2	1896. 79	-2.42	TROPO MI vs GOSAT- 2
2024-04-20	6.4337 03423	108.51 49002	TROPO MI	1895.5 1	GOSAT- 2	1896. 79	-1.27	TROPO MI vs GOSAT- 2
2024-04-20	5.9042 54913	108.40 26489	TROPO MI	1904.2 6	GOSAT- 2	1901. 8	2.45	TROPO MI vs GOSAT- 2
2024-05-01	3.4727 49472	116.58 51135	TROPO MI	1895.1 6	GOSAT- 2	1891. 84	3.32	TROPO MI vs GOSAT- 2
2024-05-07	6.7911 23867	116.76 51749	TROPO MI	1936.1 1	GOSAT- 2	1885. 9	50.21	TROPO MI vs GOSAT-

								2
2024-05-07	6.7911 23867	116.76 51749	TROPO MI	1926.1 5	GOSAT- 2	1885. 9	40.25	TROPO MI vs GOSAT- 2
2024-05-07	6.7911 23867	116.76 51749	TROPO MI	1940.6 2	GOSAT- 2	1885. 9	54.72	TROPO MI vs GOSAT- 2
2024-05-07	6.7911 23867	116.76 51749	TROPO MI	1928.0 8	GOSAT- 2	1885. 9	42.18	TROPO MI vs GOSAT- 2
2024-05-07	6.7911 23867	116.76 51749	TROPO MI	1941.1	GOSAT- 2	1885. 9	55.2	TROPO MI vs GOSAT- 2

2024-05-07	6.7911 23867	116.76 51749	TROPO MI	1931.9 6	GOSAT- 2	1885. 9	46.06	TROPO MI vs GOSAT- 2
2024-05-07	6.7911 23867	116.76 51749	TROPO MI	1938.3 9	GOSAT- 2	1885. 9	52.49	TROPO MI vs GOSAT- 2
2024-05-07	6.5021 61503	116.79 4136	TROPO MI	1930.0 4	GOSAT- 2	1889. 65	40.39	TROPO MI vs GOSAT- 2
2024-05-07	6.3830 10387	116.45 02182	TROPO MI	1922.7 6	GOSAT- 2	1891. 49	31.27	TROPO MI vs GOSAT- 2
2024-05-07	6.3830 10387	116.45 02182	TROPO MI	1927.8 9	GOSAT- 2	1891. 49	36.4	TROPO MI vs GOSAT-

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2024-05-07	6.3830 10387	116.45 02182	TROPO MI	1933.2 8	GOSAT- 2	1891. 49	41.79	TROPO MI vs GOSAT- 2
2024-05-07	6.3830 10387	116.45 02182	TROPO MI	1935.7	GOSAT- 2	1891. 49	44.21	TROPO MI vs GOSAT- 2
2024-05-07	6.3830 10387	116.45 02182	TROPO MI	1921.3	GOSAT- 2	1891. 49	29.81	TROPO MI vs GOSAT- 2
2024-05-07	6.1889 34326	116.23 62595	TROPO MI	1935.9	GOSAT- 2	1879. 87	56.03	TROPO MI vs GOSAT- 2

2024-05-07	6.1889 34326	116.23 62595	TROPO MI	1942.1 6	GOSAT- 2	1879. 87	62.29	TROPO MI vs GOSAT- 2
2024-06-10	3.0045 31384	99.674 44611	TROPO MI	1902.8	GOSAT- 2	1901. 45	1.35	TROPO MI vs GOSAT- 2
2024-06-10	3.0045 31384	99.674 44611	TROPO MI	1907.2 4	GOSAT- 2	1901. 45	5.79	TROPO MI vs GOSAT- 2
2024-06-18	5.3352 78511	116.91 98532	TROPO MI	1897.8 3	GOSAT- 2	1875. 4	22.43	TROPO MI vs GOSAT- 2
2024-06-18	5.3352 78511	116.91 98532	TROPO MI	1875.4 6	GOSAT- 2	1875. 4	0.06	TROPO MI vs GOSAT-

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2024-06-18	5.3352 78511	116.91 98532	TROPO MI	1887.0 4	GOSAT- 2	1875. 4	11.64	TROPO MI vs GOSAT- 2
2024-06-18	3.4726 6674	116.58 51746	TROPO MI	1860.5 8	GOSAT- 2	1875. 46	-14.88	TROPO MI vs GOSAT- 2
2024-06-27	2.6930 39656	102.32 85828	TROPO MI	1834.9 3	GOSAT- 2	1889. 81	-54.88	TROPO MI vs GOSAT- 2
2024-06-29	3.4725 68274	116.58 70895	TROPO MI	1870.6 6	GOSAT- 2	1862. 49	8.16	TROPO MI vs GOSAT- 2

2024-06-29	3.4725 68274	116.58 70895	TROPO MI	1890.8 7	GOSAT- 2	1862. 49	28.37	TROPO MI vs GOSAT- 2
2024-06-29	3.4725 68274	116.58 70895	TROPO MI	1880.6 2	GOSAT- 2	1862. 49	18.12	TROPO MI vs GOSAT- 2
2024-06-29	3.4725 68274	116.58 70895	TROPO MI	1890.4 3	GOSAT- 2	1862. 49	27.94	TROPO MI vs GOSAT- 2
2024-06-29	3.4725 68274	116.58 70895	TROPO MI	1864.7 2	GOSAT- 2	1862. 49	2.23	TROPO MI vs GOSAT- 2
2024-06-29	3.4725 68274	116.58 70895	TROPO MI	1875.2 4	GOSAT- 2	1862. 49	12.75	TROPO MI vs GOSAT-

								2
2024-06-29	3.4725 68274	116.58 70895	TROPO MI	1901.8	GOSAT-2	1862. 49	39.3	TROPO MI vs GOSAT-2
2024-07-05	3.4725 06523	116.58 71124	TROPO MI	1855.5 4	GOSAT-2	1865. 14	-9.6	TROPO MI vs GOSAT-2
2024-07-05	3.4725 06523	116.58 71124	TROPO MI	1866.8 7	GOSAT-2	1865. 14	1.73	TROPO MI vs GOSAT-2
2024-07-05	3.4725 06523	116.58 71124	TROPO MI	1864.6 4	GOSAT-2	1865. 14	-0.49	TROPO MI vs GOSAT-2

2024-09-28	6.7912 60242	116.76 51749	TROPO MI	1910.2 4	GOSAT- 2	1876. 91	33.33	TROPO MI vs GOSAT- 2
2024-09-28	6.7912 60242	116.76 51749	TROPO MI	1900.8 6	GOSAT- 2	1876. 91	23.95	TROPO MI vs GOSAT- 2
2024-09-28	6.7912 60242	116.76 51749	TROPO MI	1917.1	GOSAT- 2	1876. 91	40.18	TROPO MI vs GOSAT- 2
2024-09-28	6.7912 60242	116.76 51749	TROPO MI	1899.9 7	GOSAT- 2	1876. 91	23.06	TROPO MI vs GOSAT- 2
2024-09-28	6.1890 51151	116.23 62061	TROPO MI	1923.6 9	GOSAT- 2	1875. 11	48.58	TROPO MI vs GOSAT-

								2
2024-09-28	3.6745 80336	113.50 90179	TROPO MI	1868.6 6	GOSAT- 2	1897. 35	-28.7	TROPO MI vs GOSAT- 2
2024-11-01	4.4644 75155	100.79 30832	TROPO MI	1925.4 5	GOSAT- 2	1894. 22	31.22	TROPO MI vs GOSAT- 2
2024-11-01	4.2054 90589	100.71 89331	TROPO MI	1938.3 2	GOSAT- 2	1888. 98	49.34	TROPO MI vs GOSAT- 2
2024-11-01	4.2054 90589	100.71 89331	TROPO MI	1924.7 9	GOSAT- 2	1888. 98	35.81	TROPO MI vs GOSAT- 2

2024- 11-15	0.9818 54856	114.80 02777	TROPO MI	1831.4 4	GOSAT- 2	1895. 9	-64.45	TROPO MI vs GOSAT- 2
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